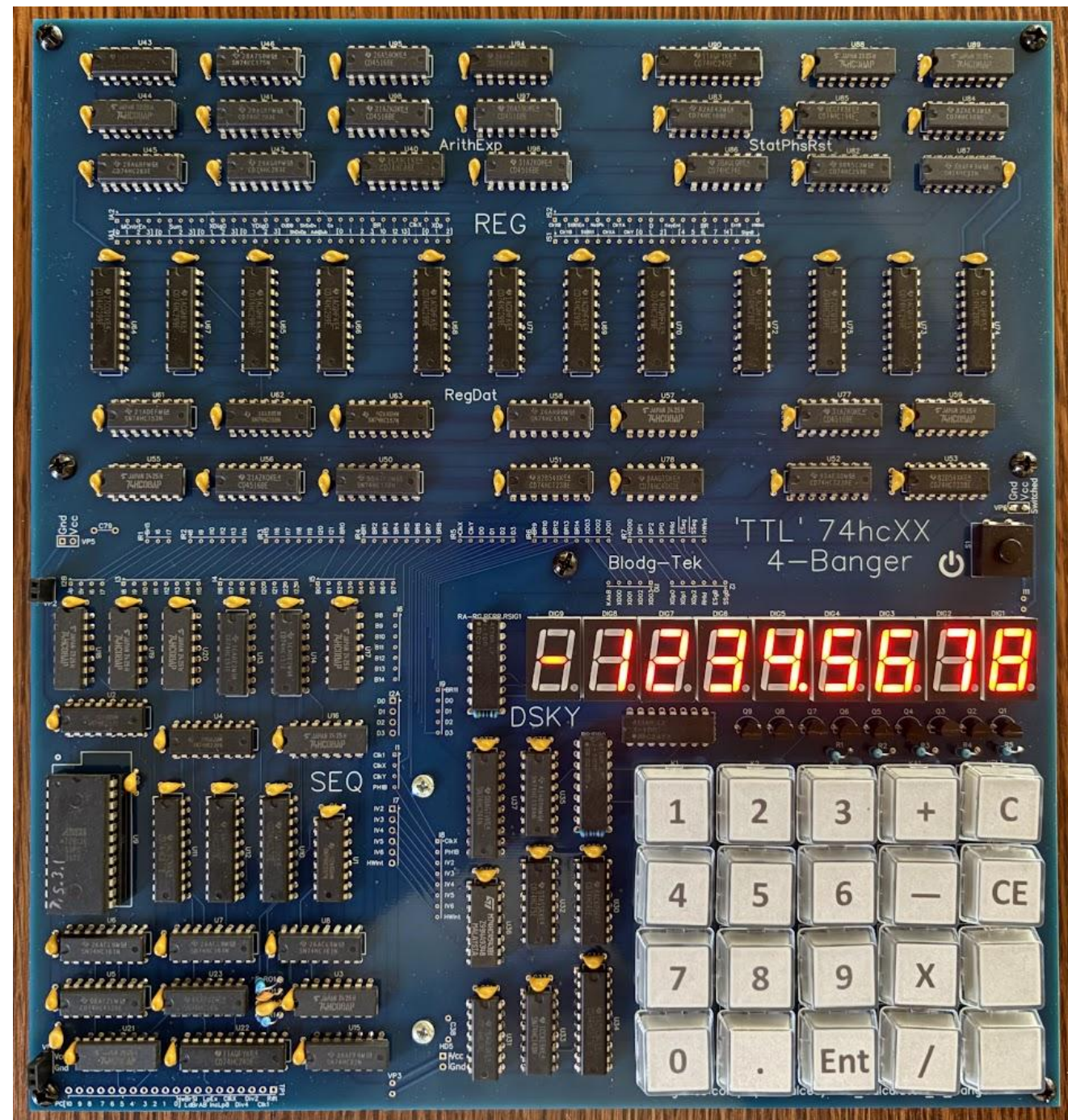


A "TTL" 4-Function Calculator Overview

Specs

- 9x10" PCB, 4 layers
- All 74hcXX logic implementation (plus 3 4XXX) chips, 77 total, and one 2816 EEPROM
- Purpose built architecture (not a CPU architecture)
- RPN style operation (HP Enter key)
- Add, subtract, multiply, and divide, clear entry, clear all
- 8 numeric digits plus an Error and sign digit
 - Max num: +99,999,999 to -99,999,999
 - Min Num: +0.0000001 to -0.0000001
- Floating decimal point
- Signals error on overflow for all operations, divide by 0, and keyboard input overflow
- Underflow handling
- Leading zero blanking during operand entry
- Running total operation (last result = Operand1 + Enter)

https://github.com/Sphaeralcea/TTL_Calculator_4_Banger



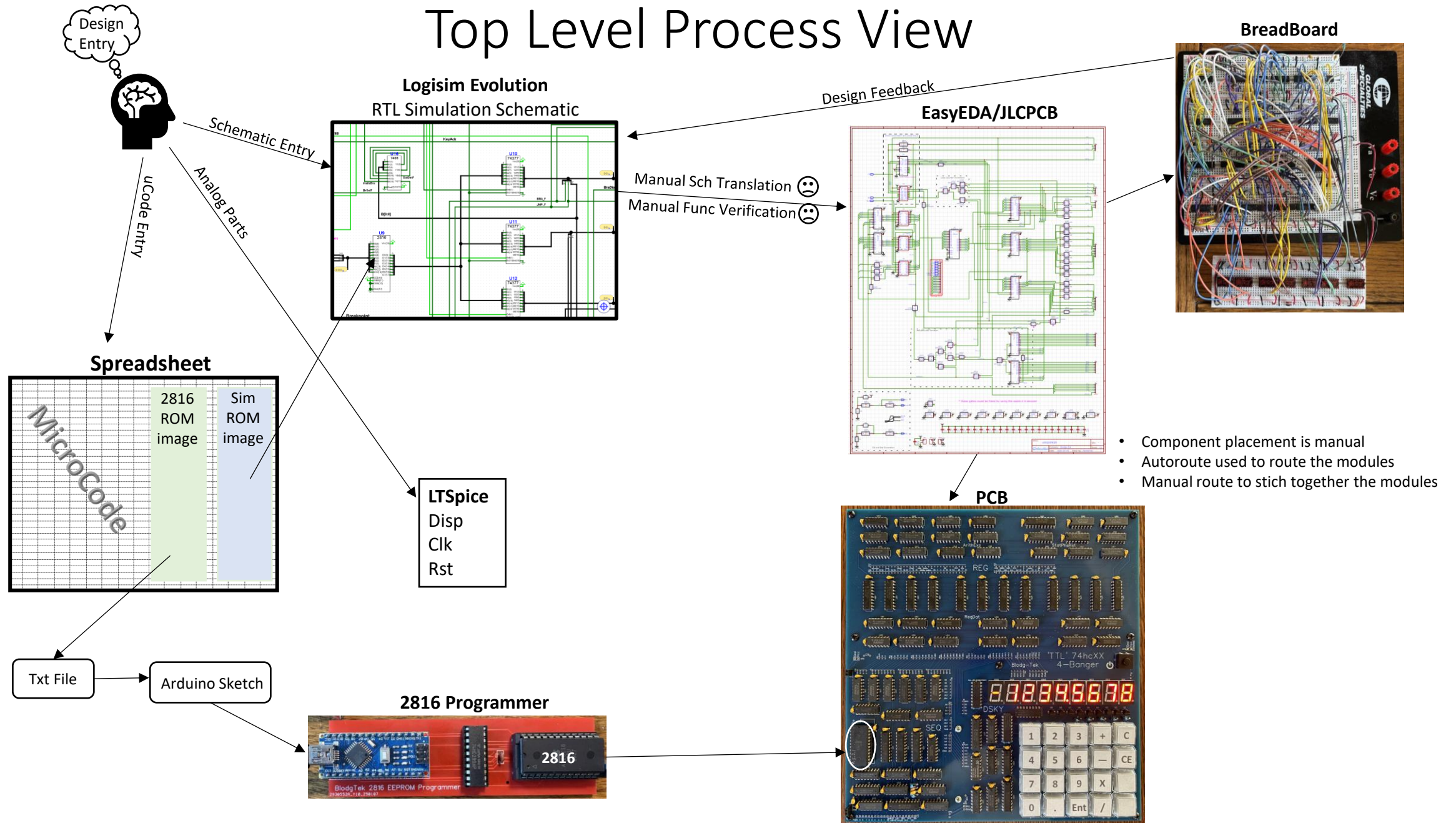
Overview

3 Major logic blocks

- SEQ: Micro-sequencer engine with 2Kx8b uROM
 - 24-bit micro-instruction word
 - Unconditional branch, 16 conditional branches, 1 NMI-type interrupt
 - Single level loop counter (no nesting)
 - Clock and reset generation
- DSKY: Keyboard and Display unit
 - 9-digit LED display sequencing,
 - 18(20)-key keyboard encoding (2 keys not used)
- REG: Register-Arithmetic unit
 - 3, 8-digit registers (4-bits X 8)
 - 1 digit BCD adder/subtractor and exponent calculator (counters)
 - 8 status and control register bits, operation progress tracking state machine

- Architectural deficits (intentional)
 - No rounding logic
 - No guard digits
- Architectural bugs (unintentional)
 - Accuracy loss dividing 2 8-digit integers
 - Underflow condition in addition not accounted for
 - No trailing zero blanking

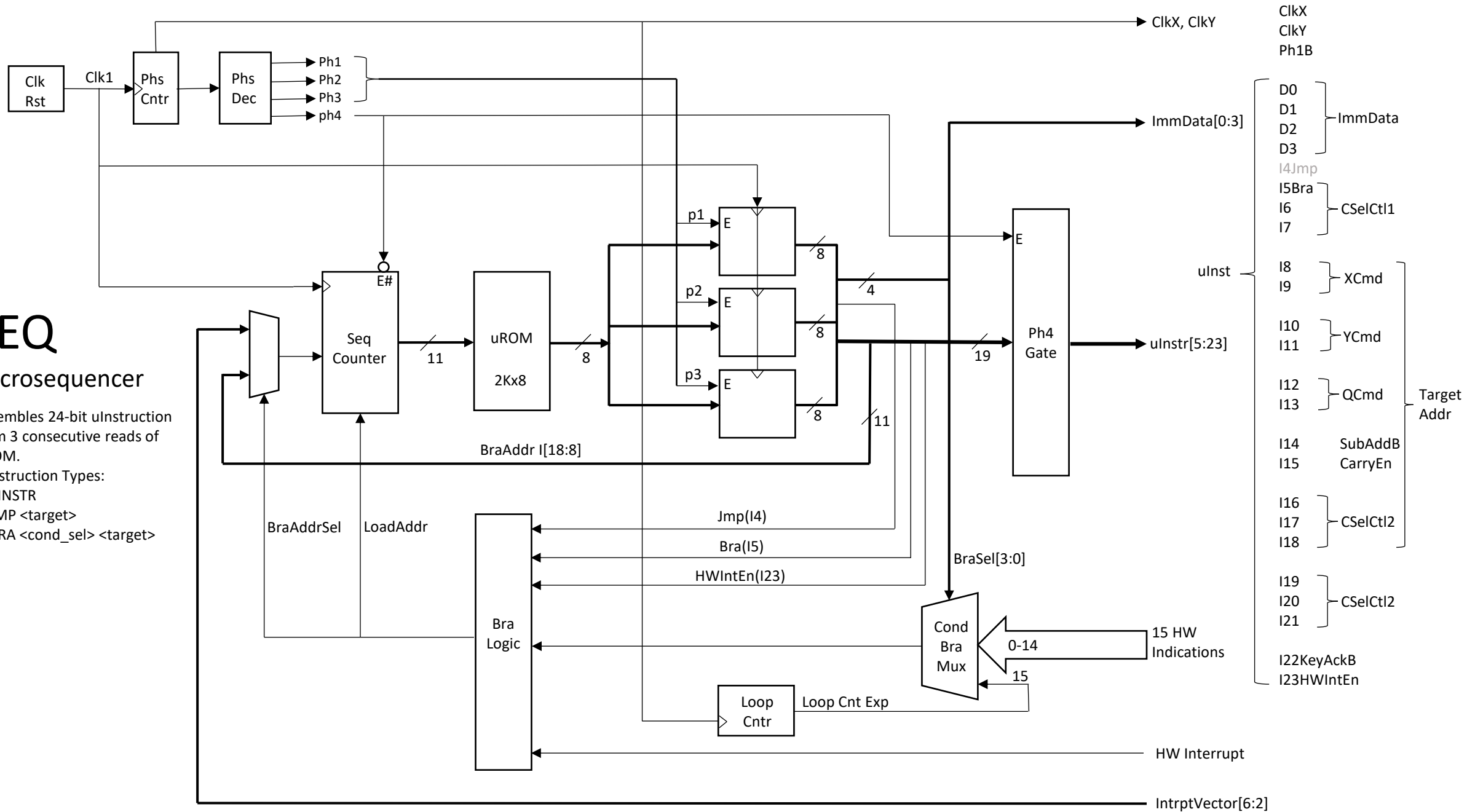
Top Level Process View



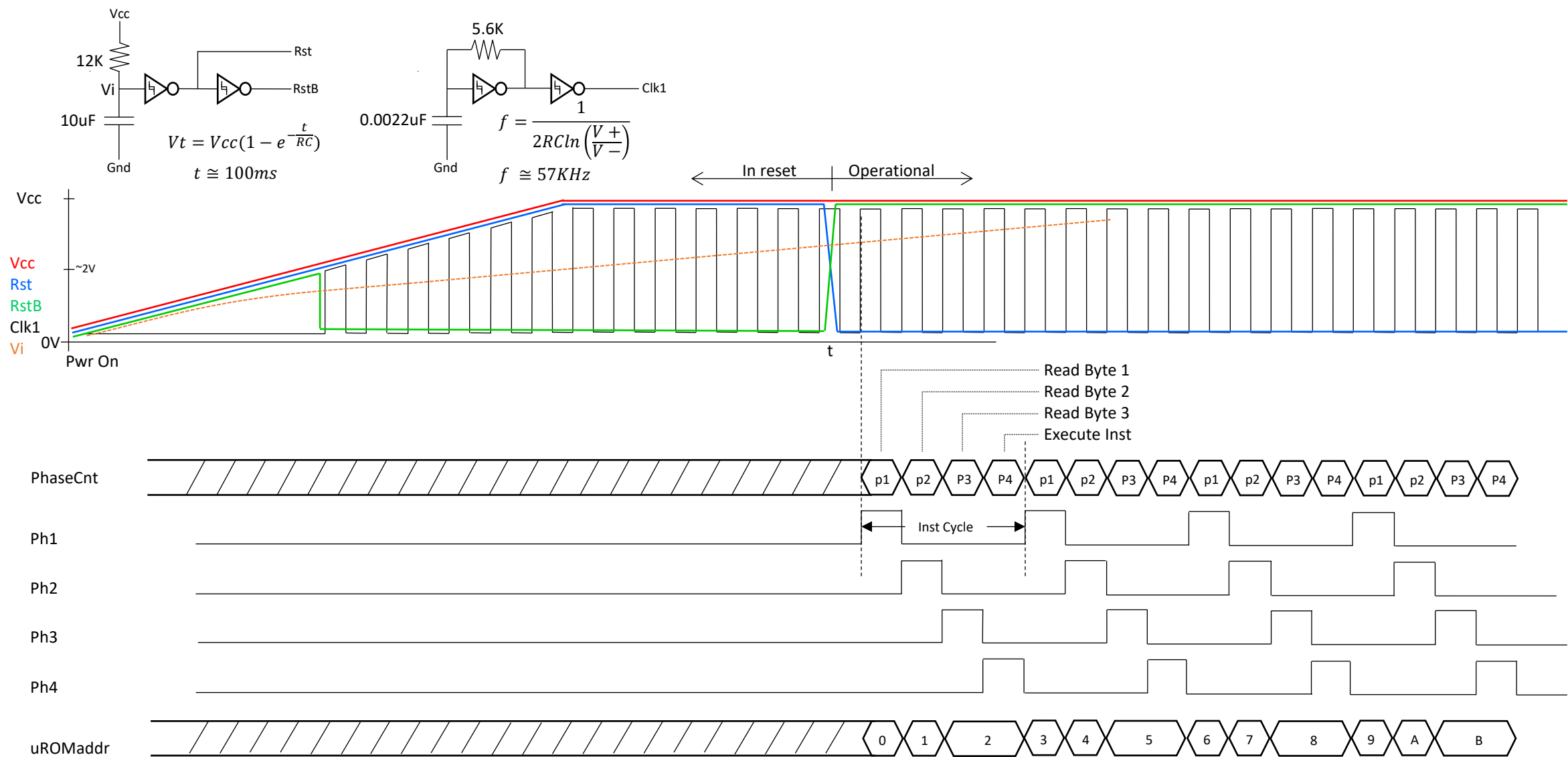
Assembles 24-bit uInstruction from 3 consecutive reads of uROM.

3 Instruction Types:

- uINSTR
- JMP <target>
- BRA <cond_sel> <target>



Power-On Reset Startup



Sequencer (SEQ)

- Includes Clock, Reset and Timing generation.
 - Schmitt trigger reset holds sequencer in reset until power rises to ~75% VCC.
 - Schmitt trigger oscillator generates 1X clock.
 - 1X clock drives SEQ logic only. Does not clock other units.
 - 1X clock feeds the uInst PC, the IR, and the phase counter which also generates the div4 clocks, ClkX and ClkY.
 - ClkX and ClkY clock other units.
- Sequencer operates on a 4-phase instruction cycle to construct and present a 23-bit micro-instruction to hardware.
 - First 3 phases read three consecutive bytes from uROM and stores them in 3 8-bit registers.
 - During first 3 phases, external hardware sees 0's (inactive) on all signals.
 - Fourth phase ungates the uInstruction and presents it to external hardware.
 - D0 – D3: Immediate data and select field
 - I4: Flow control field. This bit is reserved for the sequencer's exclusive use
 - I4=0: Entire instruction is uINSTR type
 - I4=1: Instruction is flow control (JMP or BRA)

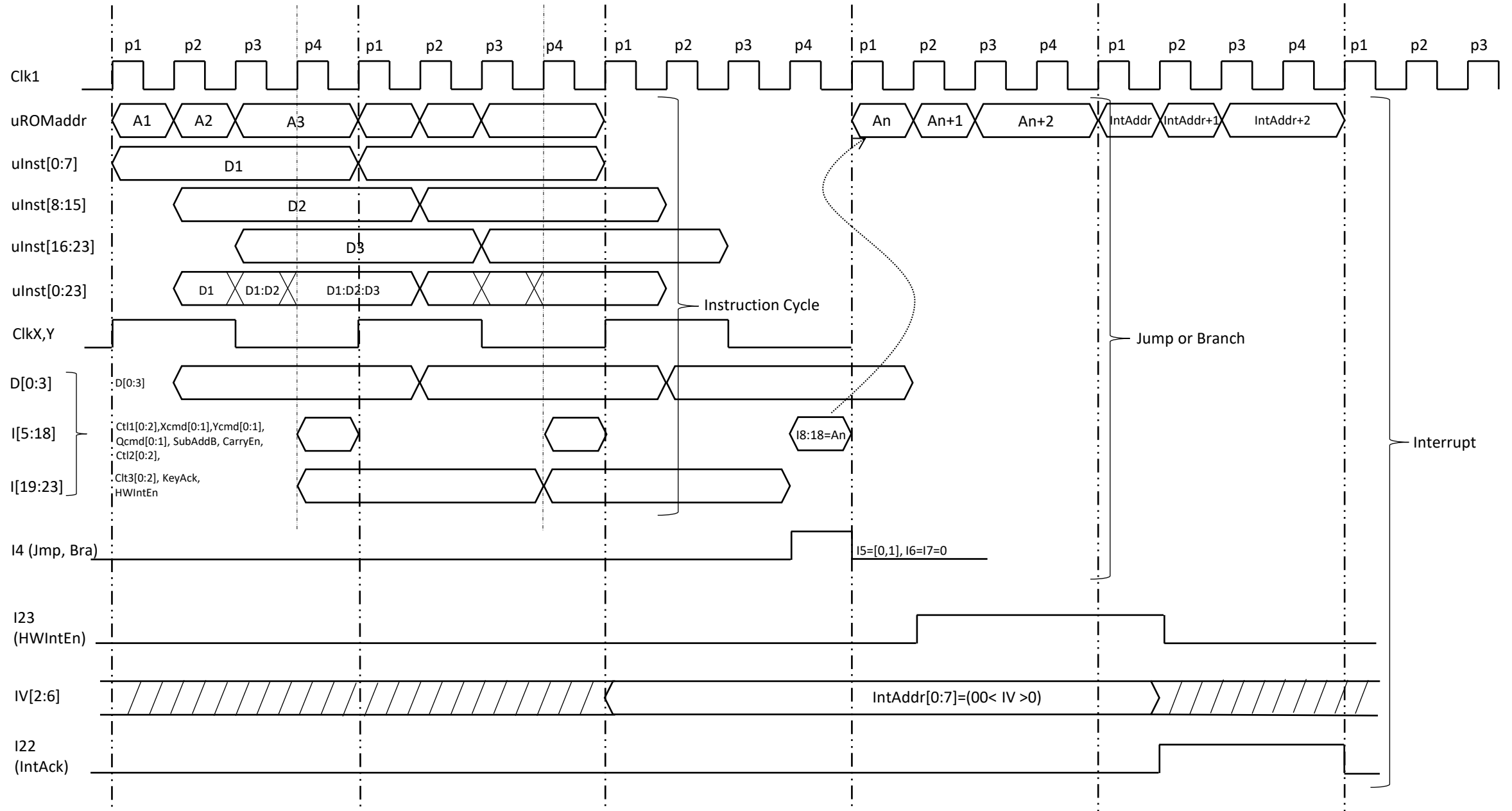
Sequencer (SEQ)

- I5: HW control and flow control qualifier.
 - If I4=0, then I5 is just another iINSTR bit
 - If I4=1, then I5 is flow control qualifier.
 - I5=0, unconditional
 - I5=1, conditional
 - One of 16 conditions may be tested, selected by D[3:0]. If condition is true then branch is taken.
 - D[3:0]=0xF is reserved. Dedicated internally to implement a simple for-loop capability.
- I6 – I21: uINSTR bits
- I22: Loop Init: Used internal to SEQ to reset a counter that implements a simple for-loop capability.
 - I22 is also used as a uINSTR bit keeping in mind its SEQ internal use
- I23: Interrupt enable. Used internal to SEQ to enable a hardware-initiated jump to be taken if external hardware has asserted the HWInt signal when I23 asserts
 - External hardware provides the target address through a 6-bit vector, IV[6:2]
 - The target address is formed by appending IV[6:2] with b00
 - Hardware jump targets are confined to uROM addresses at 4-byte boundaries between 0x10 to 0x7C
 - I23 may also be used as a uINSTR bit keeping in mind its SEQ internal use

Sequencer (SEQ)

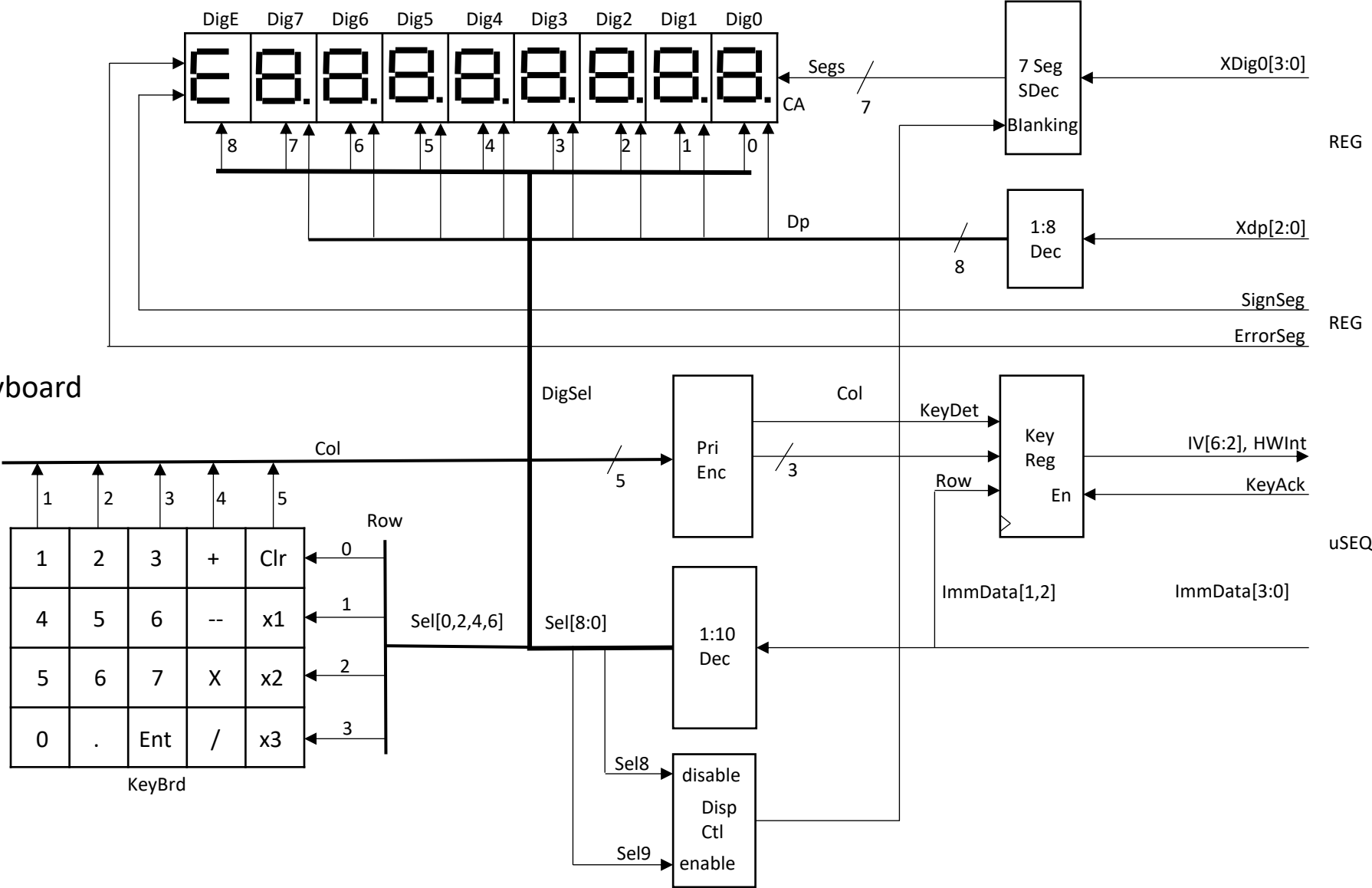
- A primitive looping structure provides the ability to create loops up to 9 iterations (0-8)
 - Loop counter reset by asserting I22 (InitLP8)
 - Loop count tested and incremented by a conditional jump with D[3:0]=0xF (BRLoop8NZ)
 - Loop count expires when loop count reaches 8 and loop will fall through on the next loop count test (BRLoop8NZ)
- Note: There is no subroutine call and return capability provided.
 - Since all keyboard service routines return to the keyboard-display loop, an unconditional JMP to this loop serves as an adequate return from keyboard service routines.

Sequencer Instruction Timing



DSKY

Display and Keyboard



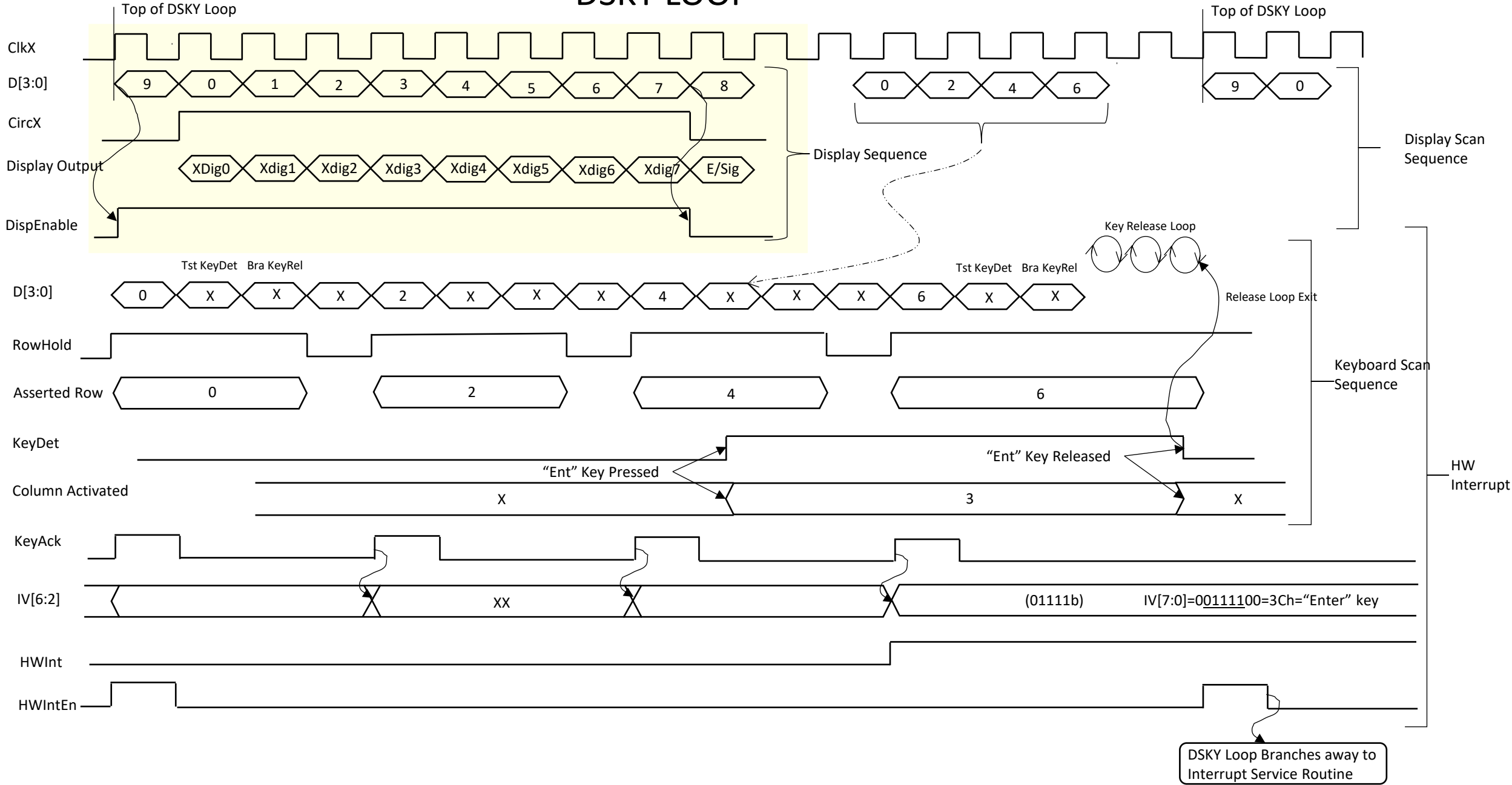
Scan Sequence

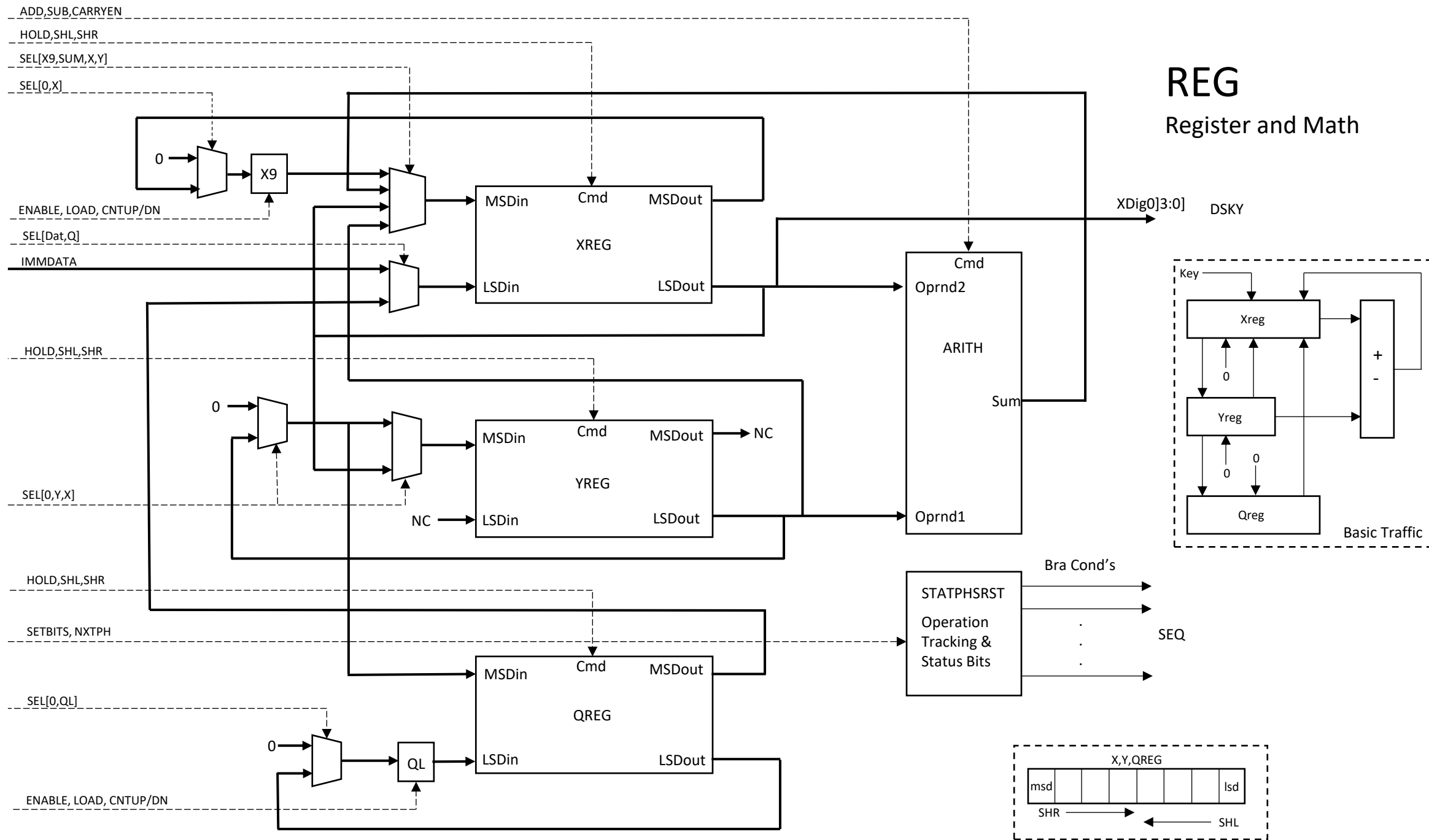
- 9 – Disp En, IntEn
- 0 – Dig0
- 1 – Dig1
- 2 – Dig2
- 3 – Dig3
- 4 – Dig4
- 5 – Dig5
- 6 – Dig6
- 7 – Dig7
- 8 – Dig8; Disp Dis
- 0 – KeyRow1, KeyAck
- 2 – KeyRow2, KeyAck
- 4 – KeyRow3, KeyAck
- 6 – KeyRow4, KeyAck

Display and Keyboard (DSKY)

- DSKY is a passive module with no autonomy. Completely controlled by SEQ
- Display and keyboard scan is performed via D[3:0] from SEQ
 - The display is enabled by D[3:0]=b1001, and it is disabled by b1000
 - Dig0 is activated by D[3:0]=b0000, Dig7 by b0111, the Sign/Error digit by b1000
 - Row 0 of the keyboard matrix is activated by b0000, row 1 by b0010, row 2 by 0100, and row 3 by b0110. If the display is enabled, digit 0, 2, 4, and 6, respectively, are activated with the keyboard row
- REG provides BCD data for each digit to display, 3-bit decimal point position, and the sign and error segments
- When a key is pressed in an activated row, a key detect signal is raised
- If microcode running on SEQ tests the detect signal to be true, then it may assert the KeyAck (I22) signal back to DSKY which registers the row and column where the depressed key resides and asserts the HWInt signal
- On any microcode instruction that asserts HWIntEn (I23) while HWInt is asserted, SEQ flow will jump to the microROM address formed by the concatenation of col[2:0], row[1:0], and b'00.

DSKY LOOP





REG: Registers, Data Steering and Compute

- REG is a passive unit with no autonomy. Completely controlled by SEQ
- Three Registers; XReg, YReg and QReg are each arrays of 8 BCD (4-bit) digits
 - Each register supports shift operations; either SHR, SHL, or both. SHR moves digits one position from MSD (most significant digit) to LSD, and the current LSD exiting the register with SRC replacing the MSD. SHL moves digits from LSD to MSD and the MSD exiting the register with SRC replacing the LSD.
 - SHR SRC: X9, SUM, X_{LSD} , Y_{LSD}
 - SHL SRC: IMMData, Q_{MSD}
 - SHRY SRC: Zero, Y_{LSD} , X_{LSD}
 - SHRQ SRC: Zero, Y_{LSD}
 - SHLQ SRC: Zero, Q_{LSD}
 - Each register may operate individually or in tandem with others but only limited cases of simultaneous register shifting are supported. Examples include: Exchange the values of Xreg and Yreg; Adding Xreg and Yreg, with SUM replacing Xreg while Yreg circulates (or is replaced with zero); Copy Xreg to Yreg while Xreg circulates (or is replaced with zero)
- X9 and QL are single BCD digit registers (loadable counters), auxiliary to Xreg and reg, respectively.
 - X9 accumulates carries out of Xmsd for Addition and Multiplication operations and is used to test Xreg digits for 0.
 - QL is used for Multiplication as a loop counter and for Division as partial quotient accumulator
 - Each can increment or decrement by one and each is loadable from two sources:
 - X9 SRC: Zero, X_{MSD}
 - QL SRC: Zero, Q_{LSD}

Arith – Add/Sub

- Arith is a single digit BCD adder/subtractor that operates only on X_{LSD} and Y_{LSD} . The output of Arith is the SUM routed only to X_{MSD}
 - Additional outputs indicate if the add generated a carry out and if the result of a subtraction is a negative number.
 - Subtraction is done through 10s complement arithmetic.
- The Add/Sub block is implemented with a cascade of three combinational logic stages:
 - 10's complement of one input
 - Summing the inputs
 - BCD correction of result
- The CarryO bit is valid for exactly one clock following the sum. This imposes a requirement for any operation on multi-digit numbers be carried out with contiguous add instructions (no intervening cycles)
- The IsNeg bit is valid for exactly one clock following the sum. This requires that action to be taken as a result of the IsNeg signal is immediate.
- If the result of a subtraction is negative, the 10's complement result is stored in Xreg and additional uCode performs subtraction of the result from 0, returning the desired result
- Arith includes 4 counters that react to key strokes to compute sum of exponents, difference of exponents, digit count, and decimal point position

Arith – Metadata Counters

- Arith includes 4 counters that react to key strokes to compute sum of exponents, difference of exponents, digit count, and decimal point position
 - Incremented or decremented by microcode instructions
 - XDigCnt – increments for each numeric key. If zero(s) is the first keystroke, it is not counted
 - XDpCnt – increments for each numeric key after the decimal point (counts digits in fractional portion of operand)
 - SexpCnt, DexpCnt – hold the sum and difference, respectively, of the exponents of the two operands
 - Positive Exponent - first key of operand is numeric. Increment for each numeric key until decimal point is keyed
 - Negative Exponent - first key of operand is decimal point key. Decrement for each leading zero keyed until non zero numeric key
 - SexpCnt
 - for each operand, positive exponents increment up, negative exponents decrement down
 - DexpCnt
 - for first operand, positive exponents increment up, negative exponents decrement down
 - For second operand, negative exponents increment up, positive exponents decrement down
- Via microcode instructions, DexpCnt can be transferred to SexpCnt. SexpCnt can be transferred to XDpCnt. DexpCnt can be transferred to XDpCnt in two steps

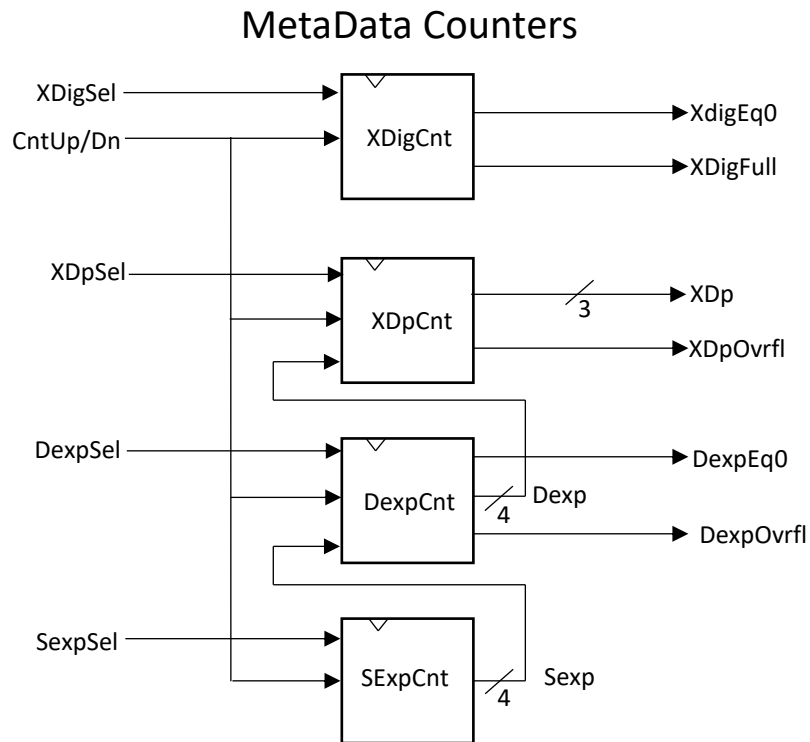
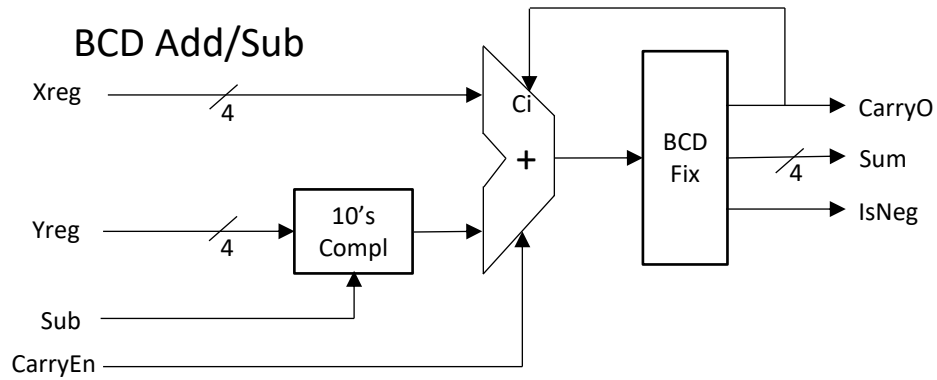
StatPhsRst

- There are two reset domains for sequential logic in REG, X and Y.
 - The Xreg domain includes Xreg and state associated with Xreg.
 - X reset is triggered by a microcode instruction and by the transition from entering the first operand to entering the second operand
 - The Yreg domain includes Yreg and Qreg and associated state.
 - Y reset is triggered by a microcode instruction
 - The Rst part of StatPhsRst is a reset synchronizer to filter spurious logic glitches from SEQ on X and Y domain resets
- The Stat part of StatPhsRst is an array of status bits to store sign bits for operands and exponents and other state information needed to parse keystrokes. Bits are set and cleared by microcode instructions and by reset events

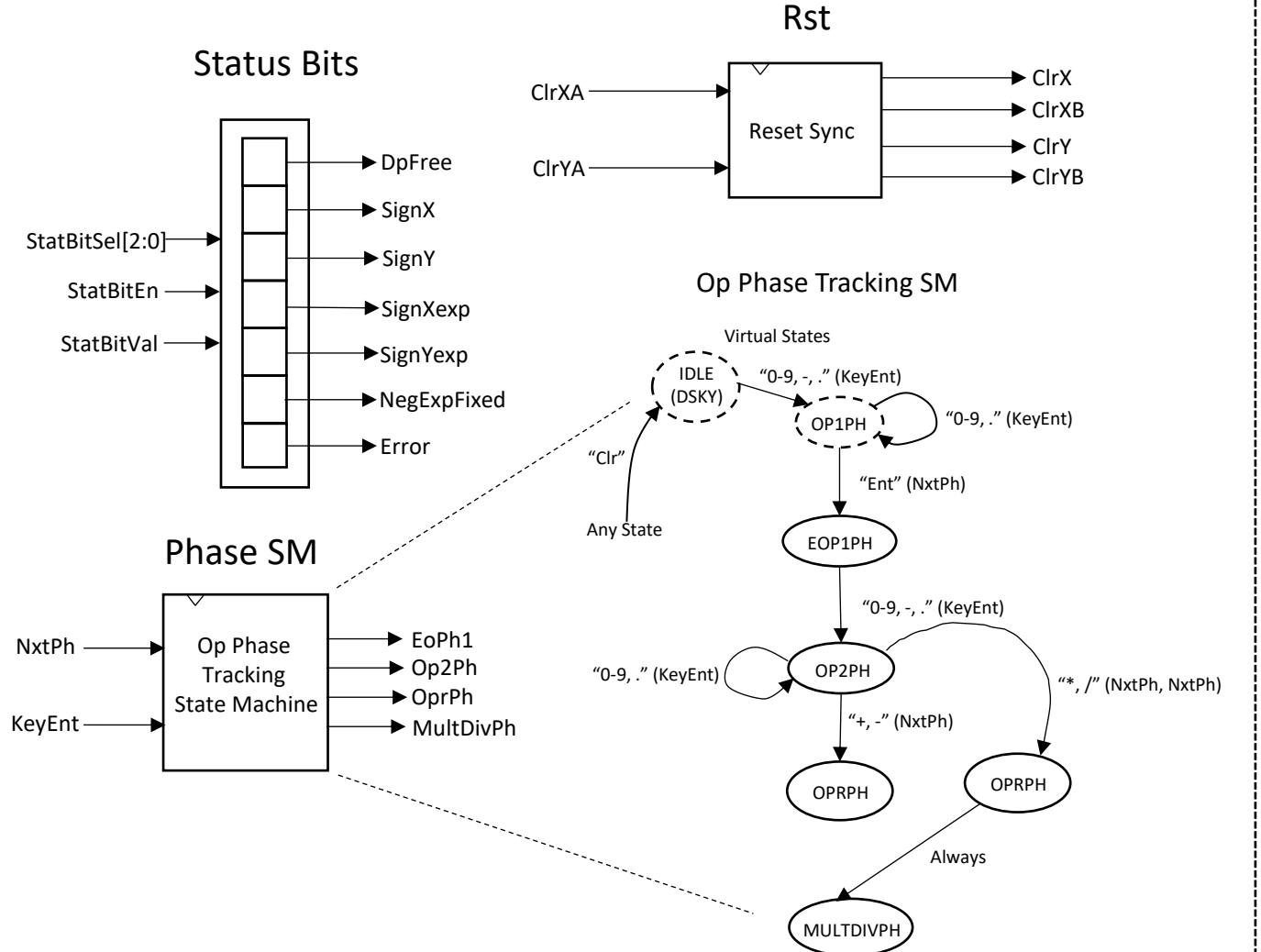
StatPhsRst

- The Phs part of StatPhsRst is a state machine that tracks the phase of the operation underway. Its state variables are tested by branch instructions
- States include
 - Idle* – System has cycle through reset but no keys have been pressed (virtual state)
 - Op1Ph* – First operand is being entered (virtual state)
 - EOp1Ph – First operand is entered but second operand has not started
 - Op2Ph – Second operand is being entered
 - OprPh – One of the 4 operations is underway
 - MultDivPh – One of either the mult or div operation is underway
- * Idle state and Op1Ph state do not need actual state variables but they represent operational states of the overall machine
- The state machine is advanced by a microcode instruction
 - The transition from EndOp1 to Op2Ph will also happen with the first key starting operand2

Arith

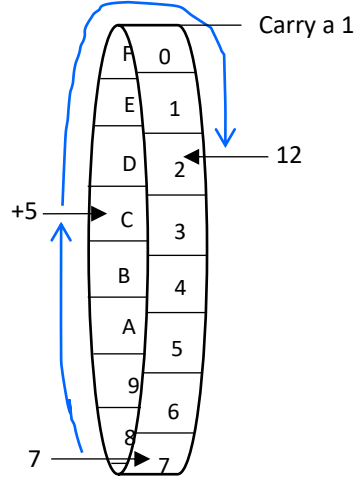


StatPhsRst



BCD Addition

Decimal addition translated to base 16 wheel:

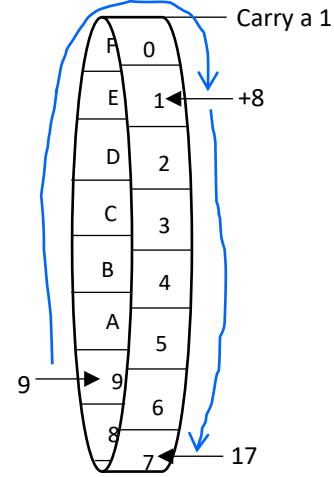


A+B; A, B in {0..9} BCD
Sum in [0..15]Hex -> [0..18]BCD

```

  7
+ 5
----
0xC   If Sum>9 Then
+ 6   Add 6 to Sum
----
12 bcd
    
```

Decimal addition translated to base 16 wheel:

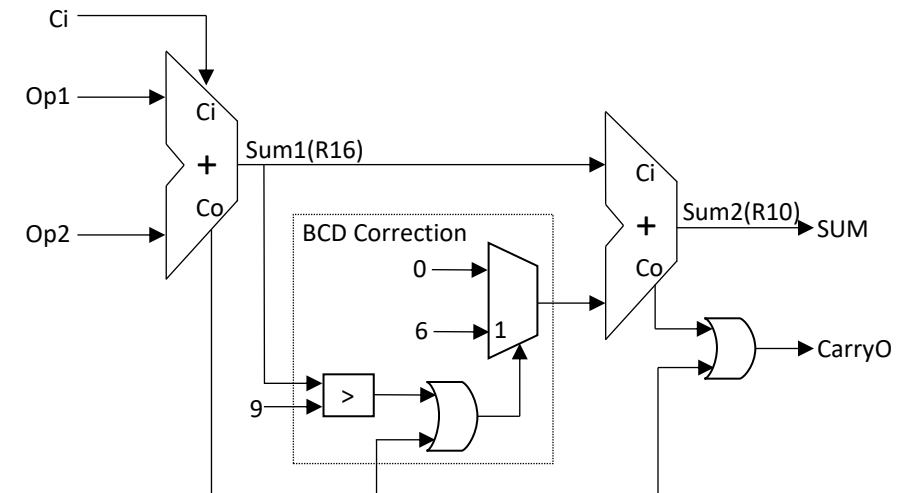


```

  9
+ 8
----
0x11  If Hex Rollover
+ 6   Add 6 to Sum
----
17 bcd
    
```

	Co		Ci	Co		Ci	Co		Ci	Co		Ci	Co	
7693		0111	1		0110	1		1001		0011				addend
+9684		+1001			+0110			+1000		+0100				addend
17377		1 0001			1101			1 0001		0111				sum1
		+0110			+0110			+0110		+0000				bdc correction
		1 0111			1 0011			0111		0111				sum2

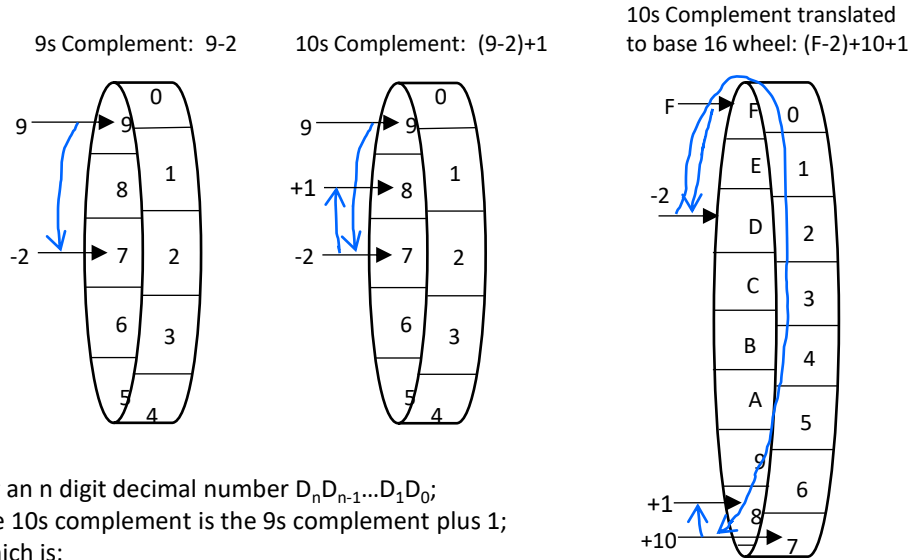
Op1+Op2+Ci	Co	Sum1	C	SUM
0	0	0	0	0
1	0	1	0	1
2	0	2	0	2
3	0	3	0	3
4	0	4	0	4
5	0	5	0	5
6	0	6	0	6
7	0	7	0	7
8	0	8	0	8
9	0	9	0	9
10	0	A	1	0
11	0	B	1	1
12	0	C	1	2
13	0	D	1	3
14	0	E	1	4
15	0	F	1	5
16	1	0	1	6
17	1	1	1	7
18	1	2	1	8
19	1	3	1	9



10's Complement

9s Complement	
0	
1	
2	←
3	
4	
5	
6	
7	←
8	
9	

↕ 9s complements



For an n digit decimal number $D_n D_{n-1} \dots D_1 D_0$;
 The 10s complement is the 9s complement plus 1;
 Which is:
 $\{9-D_i; \text{ for each } D_i (i \text{ in } 0 \text{ to } n)\} + 1$

This is the same as:
 $\{(0xF-D_i+10)\text{mod}16; \text{ for each } D_i (i \text{ in } 0 \text{ to } n)\} + 1$

This is the same as:
 $\{(1's\text{Comp}(D)+10)\text{mod}16; ; \text{ for each } D_i (i \text{ in } 0 \text{ to } n)\} + 1$

9 9 9		F	F	F
<u>-2 -9 -0</u>		<u>-2</u>	<u>-9</u>	<u>-0</u>
7 0 9	9s comp	D	6	F
<u>+ 1</u>		<u>+A</u>	<u>+A</u>	<u>+A</u>
7 1 0	10s comp	7	0	9

↔

$\left(\begin{smallmatrix} F \\ -2 \\ D \\ +A \\ 7 \end{smallmatrix} \right) \text{Mod } 16$

$\left(\begin{smallmatrix} F \\ -9 \\ 6 \\ +A \\ 0 \end{smallmatrix} \right) \text{Mod } 16$

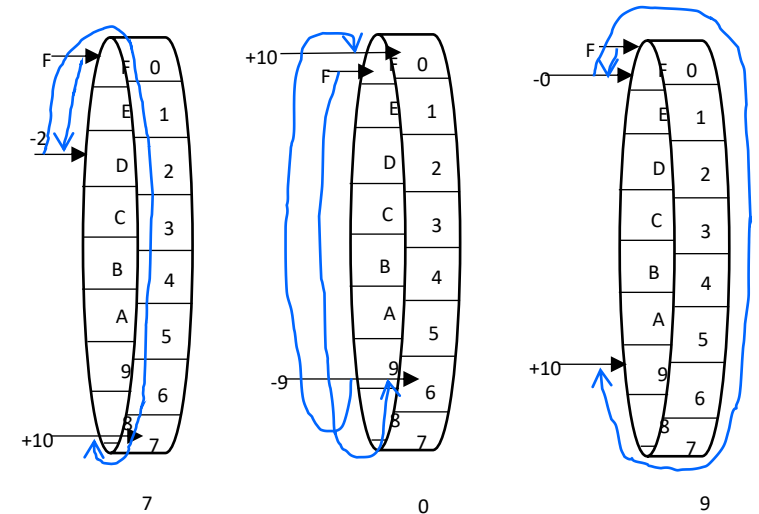
$\left(\begin{smallmatrix} F \\ -0 \\ F \\ +A \\ 9 \end{smallmatrix} \right) \text{Mod } 16$

$\begin{array}{c} + \\ 1 \end{array}$

$\begin{array}{c} 710 \end{array}$

10's complement of 290

$$9-2 = (F-2+10)\text{mod}16 = [1s\text{Comp}(2)+A]\text{mod}16 \quad F-9+10 = 1s\text{Comp}(9) + 10 \quad F-0+10 = 1s\text{Comp}(0) + 10$$



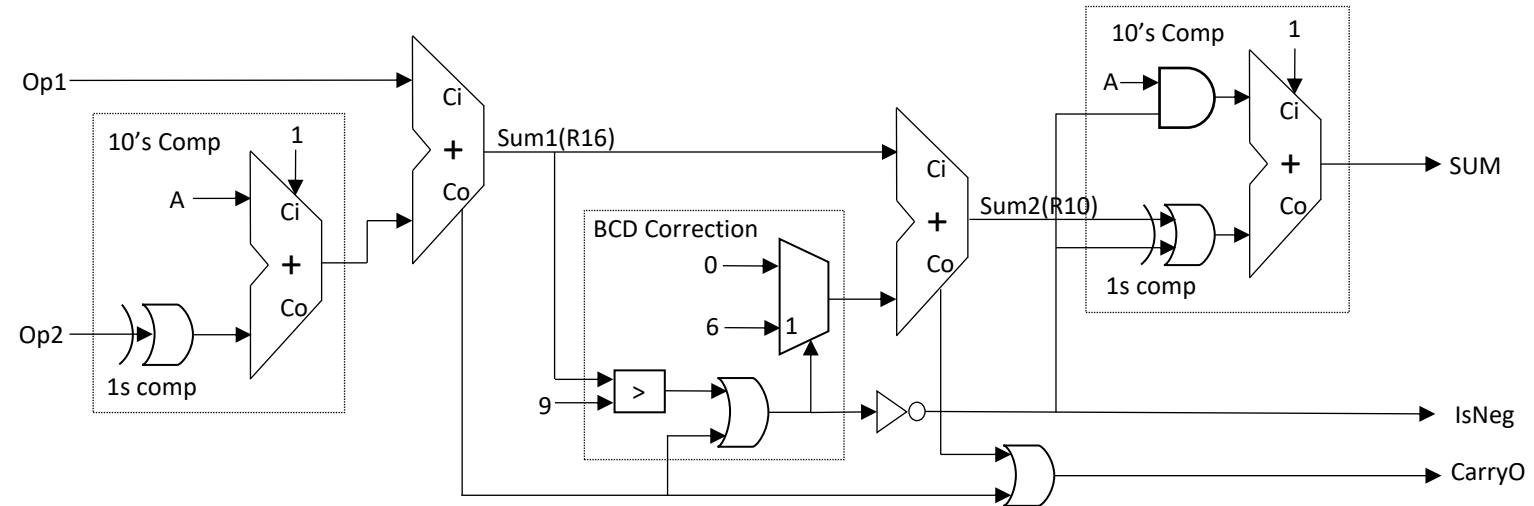
$$\begin{array}{r} 709 \\ + 1 \\ \hline 710 = 10's \text{ complement of } 290 \end{array}$$

BCD Subtraction

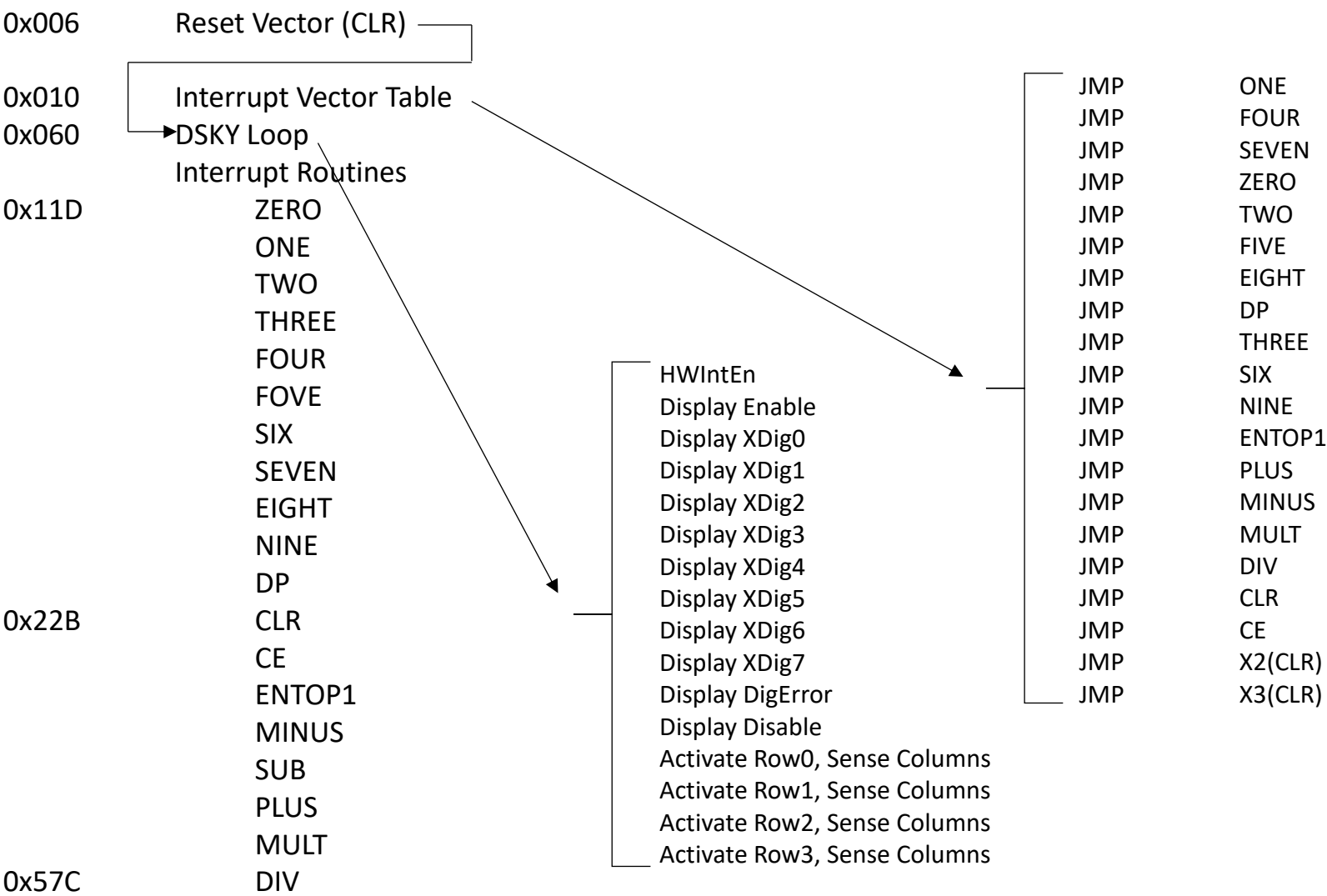
	Co	Ci	Co	Ci	
47		0100		0111	minuend
-28		-0010		-1000	subtrahend
19					
		1101		0111	1's complement of subtrahend
		+1010		+1010	add 10
	1	0111	1	0001	9's complement of subtrahend
		+0100		+0111	1 add minuend + 1
		1011		1001	difference
		+0110		+0000	bcd correction
	1	0001		1001	
Final carry=1 difference is pos					

	Co	Ci	Co	Ci	
28		0010		1000	minuend
-41		-0100		-0001	subtrahend
13					
		1011		1110	1's complement of subtrahend
		+1010		+1010	add 10
	1	0101	1	1	1000 9's complement of subtrahend
		+0010		1	+1000 1 add minuend + 1
	0	1000		0001	difference
		+0000		+0110	bcd correction
		1000		0111	
Final carry=0 difference is neg					
		0111		1000	1's complement
		+1010		+1010	add 10
		0001	1	0010	9's complement
		+0000		+0000	1 add 1
		0001		0011	

Op1-Op2	Co	A + 10#(B)	SUM
-9	0	1	-9
-8	0	2	-8
-7	0	3	-7
-6	0	4	-6
-5	0	5	-5
-4	0	6	-4
-3	0	7	-3
-2	0	8	-2
-1	0	9	-1
0	0	A	0
1	0	B	1
2	0	C	2
3	0	D	3
4	0	E	4
5	0	F	5
6	1	0	6
7	1	1	7
8	1	2	8
9	1	3	9



MicroCode Organization



MicroCode Sample

7.5.3			I[5:0], I23 1[5:0], 3[7]		I[3:0] 1[3:0]	I[22] 3[6]	I[13:8] 2[5:0]	I[7:5] 1[7:5]	I[18:16] 3[2:0]	I[21:19] 3[5:3]	I[3:0] 1[3:0]	I[14] 2[6]	I[15] 2[7]	Logisim uROM Image D[7:0] I[15:8] [23:16]					
Addr	Hex	Label	SEQ		DSKY		REG					AddSub		1[7:0]	2[7:0]	3[7:0]	4		
			Bra	Target	Sel[3:0]	KeyAck	RegCmd	SelCtl1	SelCtl2	SelCtl3	KeyMCntSel	Sub_AddB	CarryEn	v2.0 raw			Required for Logisim		
675	2A3	ENTOP1	BRSignXexp	SETOP1DPFIR										3E	A9	02	If Op1 has neg exponent (DPFirst) then save in SignYexp (		
678	2A6		JMP	TSTDIGEMPTY										10	AC	02	Otherwise goto test for empty X reg		
681	2A9	SETOP1DPFIR							StatBitEn	StatBit1	SignYexp			05	00	1B			
684	2AC	TSTDIGEMPTY	BRDigEmpty	ENTERZERO										32	B5	02	If no digit hit yet, Jump to enter0		
687	2AF		BRDPFree	ADJEXPFIXDP										37	BE	02	If DP was never hit then goto adjust exp and fixDP		
690	2B2		JMP	LEFTJUST										10	C7	02	Otherwise go to left justify		
693	2B5	ENTERZERO							CntrEn	Cntrlncr	XDigCnt			01	00	36	Incr Xreg Dig count (enter a 0)		
696	2B8								StatBitEn	StatBit1	FixDP			00	00	1B	and fix the DP		
699	2BB		JMP	LEFTJUST										10	C7	02	then left justify		
702	2BE	ADJEXPFIXDP							CntrEn		XYSexpCnt			04	00	06	If X is whole integer, fix DP and adjust exponent		
705	2C1								CntrEn		XYDexpCnt			03	00	06			
708	2C4								StatBitEn	StatBit1	FixDP			00	00	1B			
711	2C7	LEFTJUST	BRDigFull	COPYXY										31	D6	02	Left justify X until Xdig is full		
714	2CA						LeftJustX	SelXData			0			00	02	00	Shift left with 0's in at msd		
717	2CD								CntrEn	Cntrlncr	XDigCnt			01	00	36	and adjust Xdp and Xdig count to match		
720	2D0								CntrEn	Cntrlncr	XDPCnt			02	00	36			
723	2D3		JMP	LEFTJUST										10	C7	02			
726	2D6	COPYXY				InitLp8								00	00	40	Copy Xreg to Yreg		
729	2D9		BRLoop8NZ	COPYXYE										3F	DC	02			
732	2DC	COPYXYE					CopyXY	SelXReg	SelXXlsl	SelYXlsl				20	05	28			
735	2DF		BRLoop8NZ	COPYXYE										3F	DC	02			
738	2E2		BRSignX	CPSIGXSIGY										34	E8	02	Copy Xsign to Ysign		
741	2E5		JMP	RMLDZEROS										10	EB	02			
744	2E8	CPSIGXSIGY							StatBitEn	StatBit1	SignY			04	00	1B			
747	2EB	RMLDZEROS				InitLp8								00	00	40	Loop to remove leading zeros from fractional args. Clea		
750	2EE	BRN1E	BRLoop8NZ	RLZENTERE										3F	F4	02	Burn loop 1 of 9 and jump to entry point		
753	2F1	RLZTOPE					RmLdZeroX	SelXData			0			00	02	00	Shift Xmsd out the top and zero in from the bottom		

MicroCode Fields, Mnemonics and Codes

	Jmp/Bra	1[4]	1[3:0]	RegCmd	2[5:0]	SelCtl1	1[7:5]	SelCtl2	3[2:0]	CtlSel3	3[5:3]	KeyMCntSel	1[3:0]
	JMP	1	0000	CircX	000001	SelXReg	001	ClrXPh1	001	SelQLQlSd	001	0	0000
BR0	BRDecEven	1	0000	DPAdjX	000001	LoadX9	010	ClrYPh2	010	SelYQYlSd	010	1	0001
BR1	BRDigFull	1	0001	EntNumX	000010	SelXQmsd	011	StatBitEn	011	StatBit1	011	2	0010
BR2	BRDigEmpty	1	0010	LeftJustX	000010	LoadQL	100	SelXYlSd	100	SelX9Xmsd	100	3	0011
BR3	BRDintOvrfl	1	0011	DPAdjY	000100	SHDexpXDP	101	SelXSum	100	SelXYlSd	101	4	0100
BR4	BRSignX	1	0100	CopyXY	000101	Reserved1	110	NxtPh	101	CntrlIncr	110	5	0101
BR5	BRSignY	1	0101	ExchXY	000101	Reserved2	111	CntrEn	110	RowHold	111	6	0110
BR6	BRInOp2Ph	1	0110	AddSub	000101	HWINTEN	000	SHSexpDexp	111	SelYZero	000	7	0111
BR7	BRDPFree	1	0111	DivXY10	000101	SelXData	000	SelXX9	000	SelQZero	000	8	1000
BR8	BRQLZero	1	1000	ShrX9X8QDiv10	010001	SelXSumX9	000	SelXXlSd	000	SelX9Zero	000	9	1001
BR9	BRX9Zero	1	1001	CopyYQ	010100	ExtraSelCtl1	000	ExtraSelCtl1	000	SelQLZero	000	BlankNum	1111
BR10	BRDpOvrfl	1	1010	CopyQX	100010	ExtraSelCtl2	000	ExtraSelCtl2	000	ExtraSelCtl3	000	XDigCnt	0001
BR11	BRNoKeyDet	1	1011	ApendDivdn	000010	ExtraSelCtl3	000					XDPCnt	0010
BR12	BRIsNeg	1	1100	RmLdZeroX	000010	ExtraSelCtl4	000	ALUCmd	2[7:6]	SEQ/DSKY	3[7:6]	XYDexpCnt	0011
BR13	BRCarryO	1	1101	CopyQ0QL	100000			SubXY	x1	HWINTEN	10	XYSexpCnt	0100
BR14	BRSignXexp	1	1110	ShrX	000001	BR*	001	CarryEn	1x	KeyAck	01	X9Cnt	0101
BR15	BRLoop8NZ	1	1111	ShlX	000010					InitLP8	01	QLCnt	0110
	BRSignYexp	1	0110	CopyYX	000101							FixDP	0000
	BrNegExpTerm	1	0101	ExtraCmd3	000000							SignXexp	0001
	ExtraBR2	0	0000									NegExpTerm	0010
	ExtraBR3	0	0000									SignX	0011
	HWINTEN	0	0000									SignY	0100

ImmData asserts on Ph2 of each cycle

SelCtl1, SelCtl2, RegCmd, ALUCmd assert on Ph4 and is valid only Ph4

SelCtl3, KeyAckB, HWIntEn Assert on Ph4 of each cycle

SelCtl3 Can assert simutaneous with JMP/BR*. SelCtl1/2 cannot.

MicroInstruction Fields

SelCtl1 (I5br:I7)

SelXReg	001	Select X reg MSD source from X reg or Y reg group
LoadX9	010	Load X9 source into X9 reg
SelXQmsd	011	Select X reg source from Q Dig7
LoadQL	100	Load QL source into QL reg
SHDexpXDP	101	Load Dexp counter value into XDP counter
Reserved1	110	Currently unassigned
Reserved2	111	Currently unassigned
SelXData	000	Select X LSD reg source from D[3:0]
SelXSumX9	000	Select X reg MSD source from Sum or X9 group

SelCtl2 (I16:I18)

SelQLQlsd	001	Select QL reg source from Q reg LSD
SelYQYlsd	010	Select Y and Q reg MSD sources from Y reg LSD
StatBit1	011	Set selected status bit to 1
SelX9Xmsd	100	Select X9 reg source from X reg MSD
SelYXlsd	101	Select Y reg source from X reg LSD
Cntrlncr	110	Increment selected and enabled counter
RowHold	111	DSKY row latch disable (hold)
SelYZero	000	Select Y reg MSD source from 0
SelQZero	000	Select Q reg MSD source from 0
SelX9Zero	000	Select X9 reg source from 0
SelQLZero	000	Select QL reg source from 0

SelCtl3 (I19:I21)

ClrXPh1	001	Clear X reg domain devices
ClrYPh2	010	Clear Y reg domain devices
StatBitEn	011	Enable status latches for write
SelXYlsd	100	Select X reg MSD source from Y reg LSD
SelXSum	100	Select X reg MSD source from SUM
NxtPh	101	Advance tracking state machine state
CntrlEn	110	Enable selected counter for up/down advance
SHSexpDexp	111	Load Sexp counter value into Dexp
SelXX9	000	Select X reg MSD source from X9 reg
SelXXlsd	000	Select X reg MSD source from X reg LSD

RegCmd (I8:I13)

CircX	000001	Shift X reg from MSD toward LSD
DPAdjX	000001	
ShrX	000001	
EntNumX	000010	Shift X reg from LSD toward MSD
LeftJustX	000010	
ApendDivdn	000010	
RmLdZeroX	000010	
DPAdjY	000100	Shift X reg from MSD toward LSD
CopyXY	000101	Shift X and Y reg from MSD toward LSD
ExchXY	000101	
AddSub	000101	
DivXY10	000101	
ShrX9X8QDiv10	010001	Shift X and Q reg from MSD toward LSD
CopyYQ	010100	Shift Y and Q reg from MSD toward LSD
CopyQ0QL	100000	Shift Q reg from LSD toward MSD
CopyQX	100010	Shift X and Q reg from LSD toward MSD

Branches (D3:D0)

JMP	NA	Unconditional Branch
BRDecEven	0000	Branch if DexpCnt=0
BRDigFull	0001	Branch of DigCnt=8
BRDigEmpty	0010	Branch if DigCnt=0
BRDintOvrfl	0011	Branch if Dexp>=8
BRSignX	0100	Branch if SignX status bit set
BRSignY	0101	Branch if SignY status bit set
BRInOp2Ph	0110	Branch if in or beyond Op2Ph tracking state
BRDPFree	0111	Branch of DP status bit has not been set
BRQLZero	1000	Branch if Qlreg=0
BRX9Zero	1001	Branch if X9reg=0
BRDpOvrfl	1010	Branch if XDpCnt>=8
BRNoKeyDet	1011	Branch if no key is currently pressed
BRIsNeg	1100	Branch if arithmetic (SUM) sign bit is set
BRCarryO	1101	Branch if arithmetic carry out bit is set
BRSignXexp	1110	Branch if Op2's exponent sign status bit is set
BRLoop8NZ	1111	Branch if LoopCnt>0
BRSignYexp	0110	Branch if Op1's exponent sign status bit is set
BrNegExpTerm	0101	Branch if current negative exponent calculated

Immediate Data Labels (D3:D0)

0	0000	Immediate data 0
1	0001	Immediate data 1
2	0010	Immediate data 2
3	0011	Immediate data 3
4	0100	Immediate data 4
5	0101	Immediate data 5
6	0110	Immediate data 6
7	0111	Immediate data 7
8	1000	Immediate data 8
9	1001	Immediate data 9
XDigCnt	0001	Select the XDigit counter
XDPCnt	0010	Select the XDP counter
XYDexpCnt	0011	Select the Dexp counter
XYSexpCnt	0100	Select the Sexp counter
X9Cnt	0101	Select the X9 counter
QLCnt	0110	Select the QL counter
FixDP	0000	Select the DP status bit
SignXexp	0001	Select the Op2 exponent sign status bit
NegExpTerm	0010	Select the negative exponent accumulation terminated status bit
SignX	0011	Select the Op2 sign status bit
SignY	0100	Select the Op1 sign status bit
SignYexp	0101	Select the Op1 exponent sign status bit
AssertError	0110	Select the Error status bit

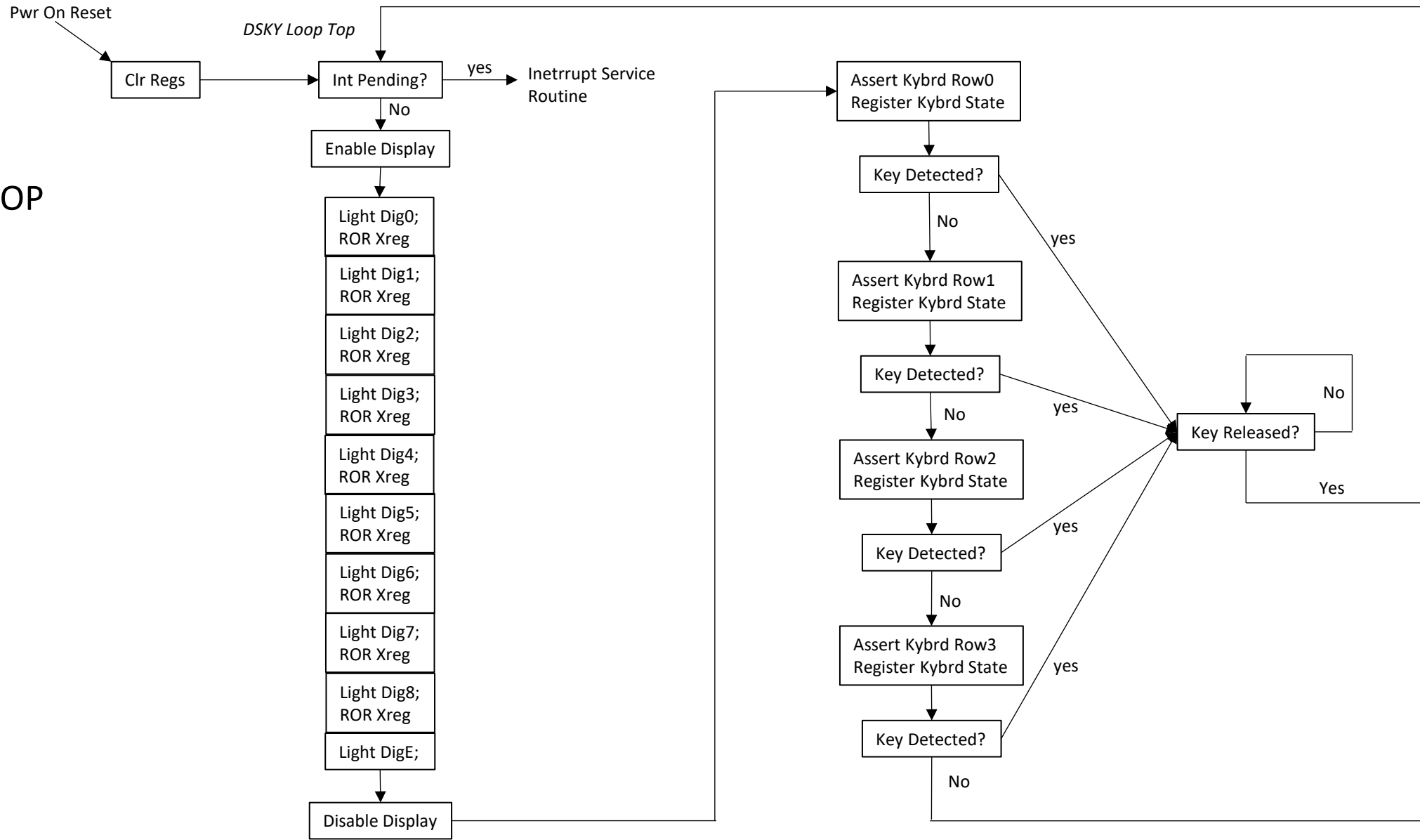
Add/Sub (I14:I15)

SubAddB	0	Add Digit 0 of Y register with Digit 0 of X register
	1	Subtract Digit 0 of X register from Digit 0 of Y register
CarryEn	0	Disable Carry In to Adder/Subtractor for one instruction
	1	Enable Carry In to Adder/Subtractor for one instruction

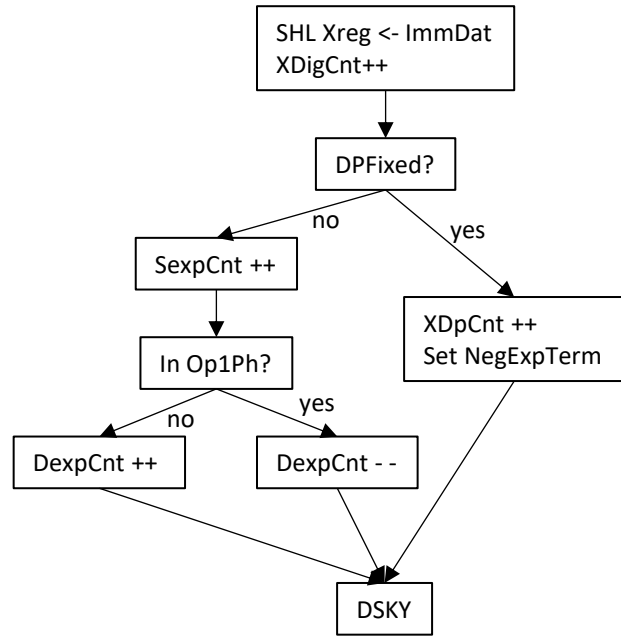
SEQ/DSKY (I22:I23)

KeyAck	0	NA
	1	Write state of keyboard into keyboard register, reset sequencer loop counter
HWIntEn	0	NA
	1	Enable branch to interrupt service routine if HWInt is active for one instruction

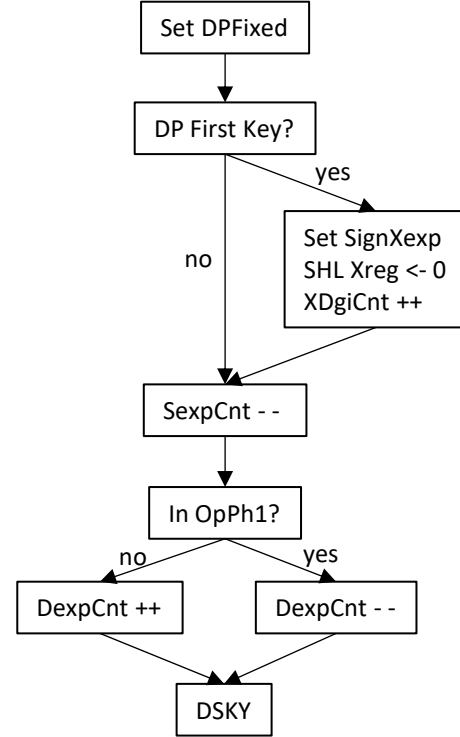
DSKY LOOP



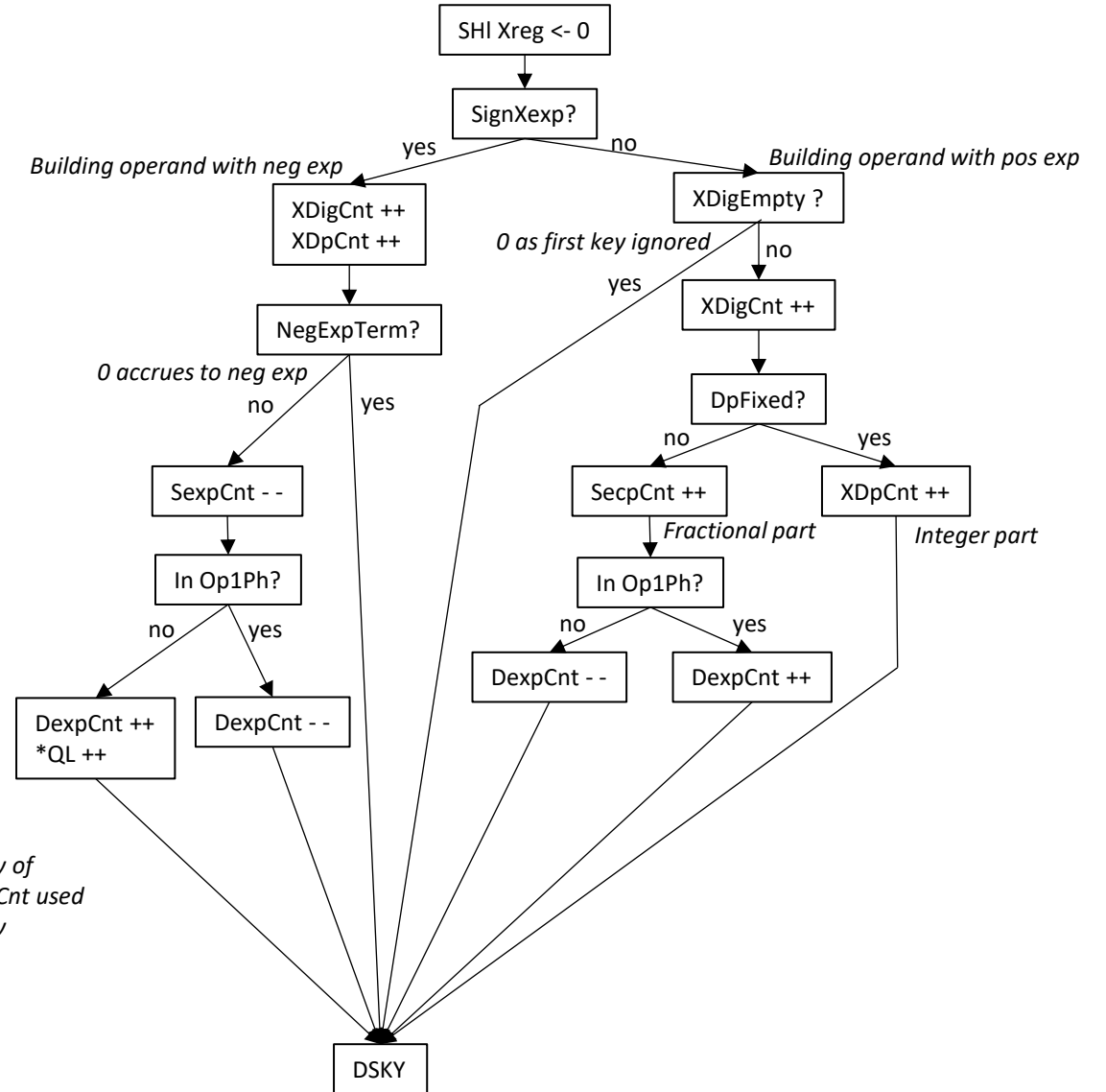
1-9 Service Routine



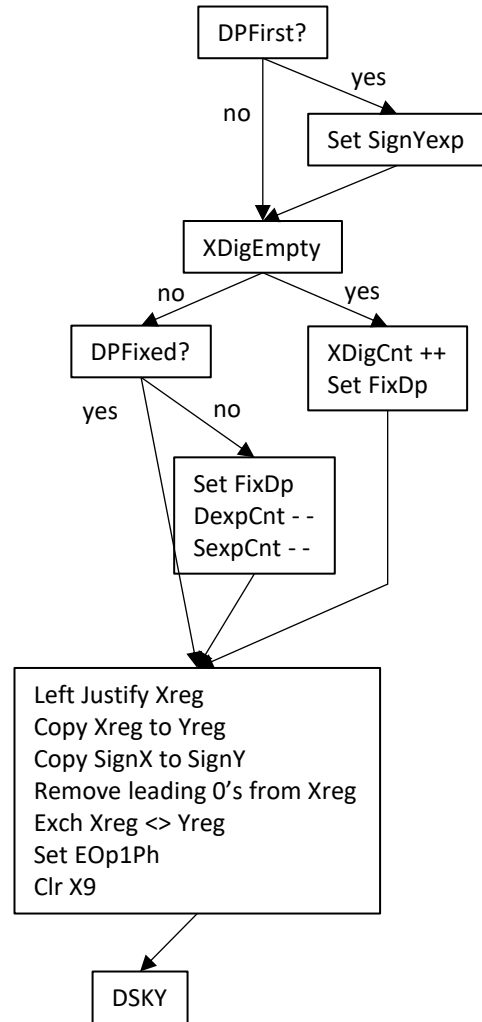
Dp Service Routine



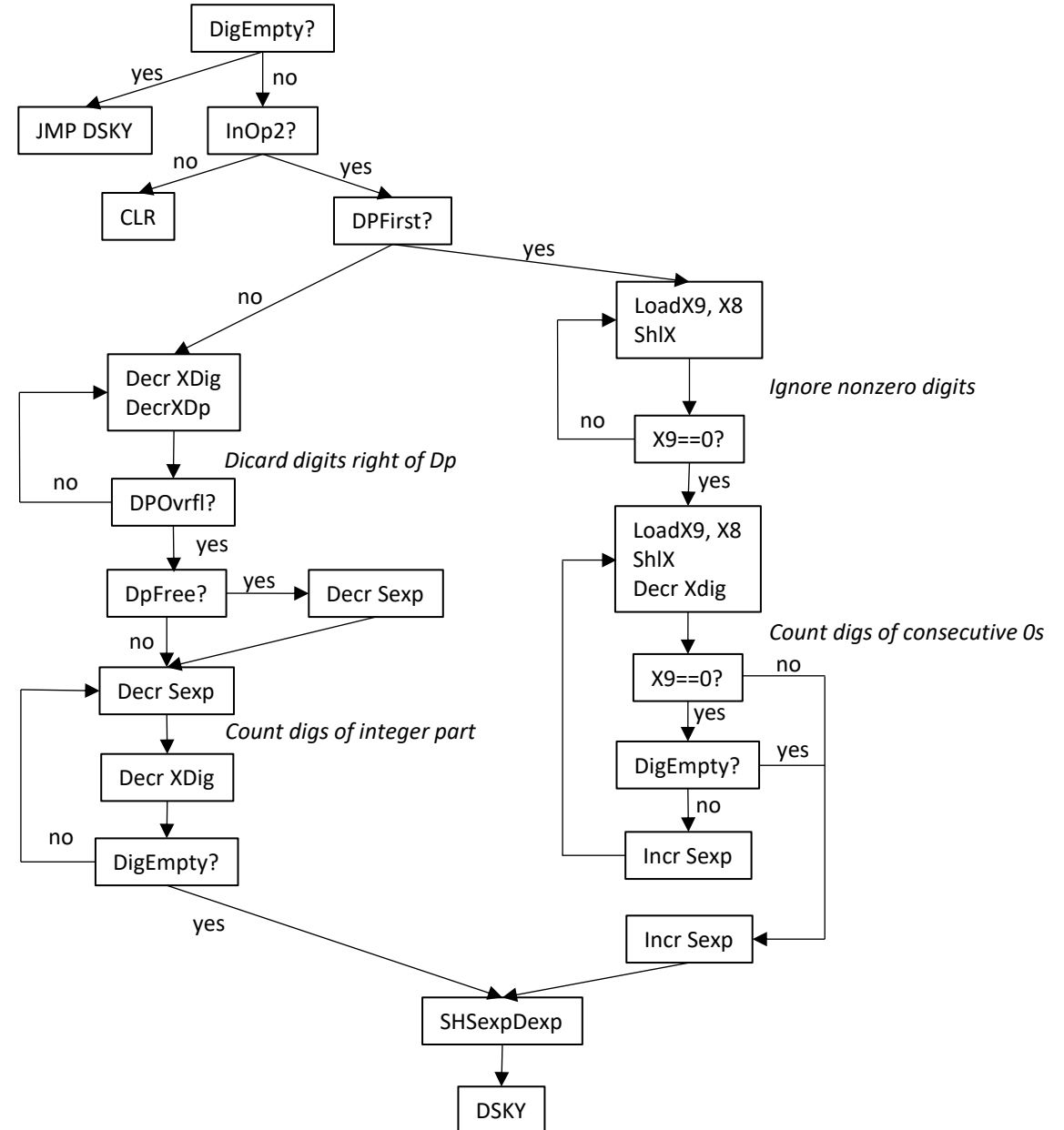
Zero Service Routine



Enter Op1 Service Routine



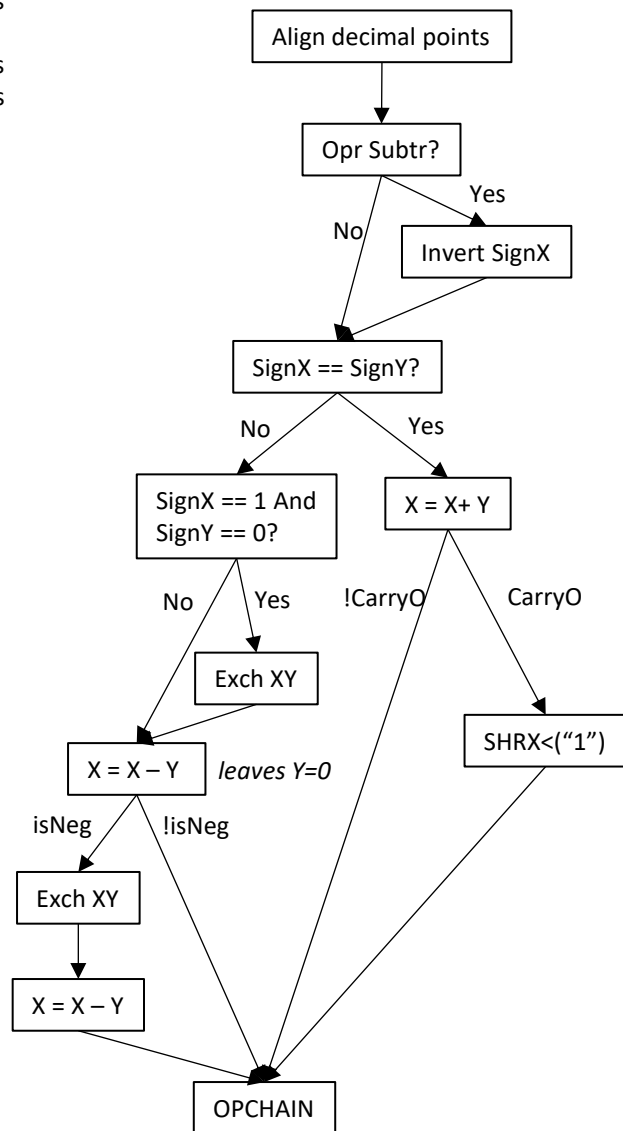
CE Service Routine



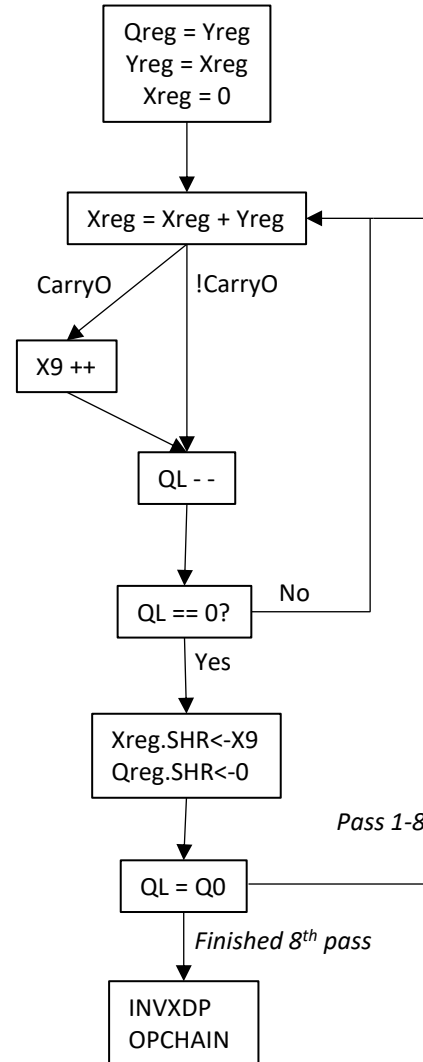
Basic Register Model

Xreg : Array[8:1] of BCD digits
X9 : BCD digit
Yreg : Array[8:1] of BCD digits
Qreg : Array[8:1] of BCD digits
QL : BCD digit
Xdp : Integer[2:0]
DpFixed : Boolean
Sexp : Integer[3:0]
Dexp : Integer[3:0]
NegExpDpFixed : Boolean

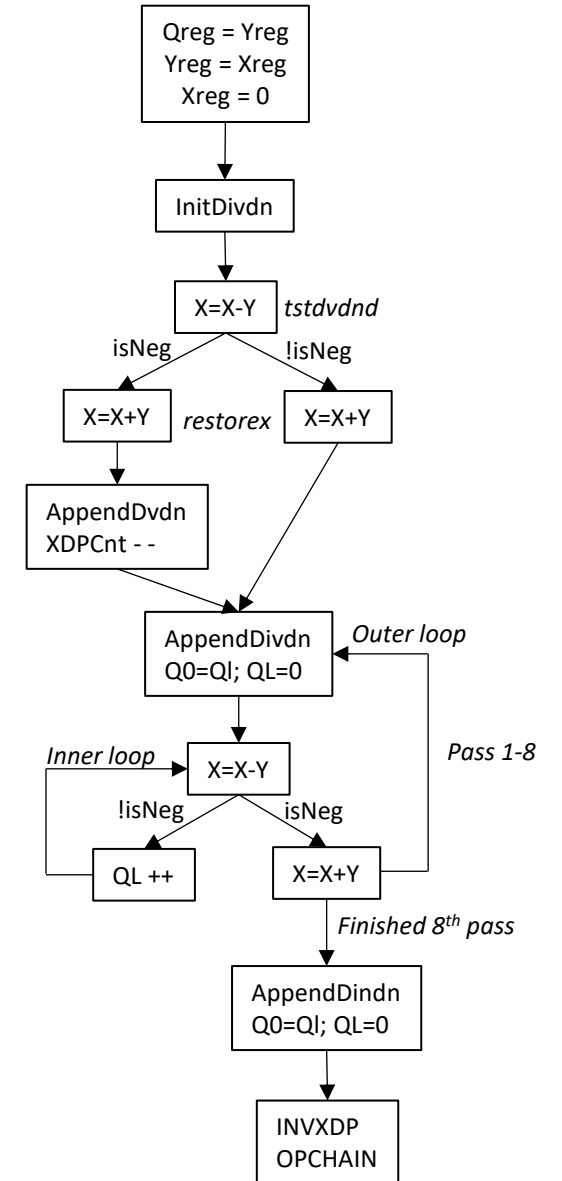
Add or Sub



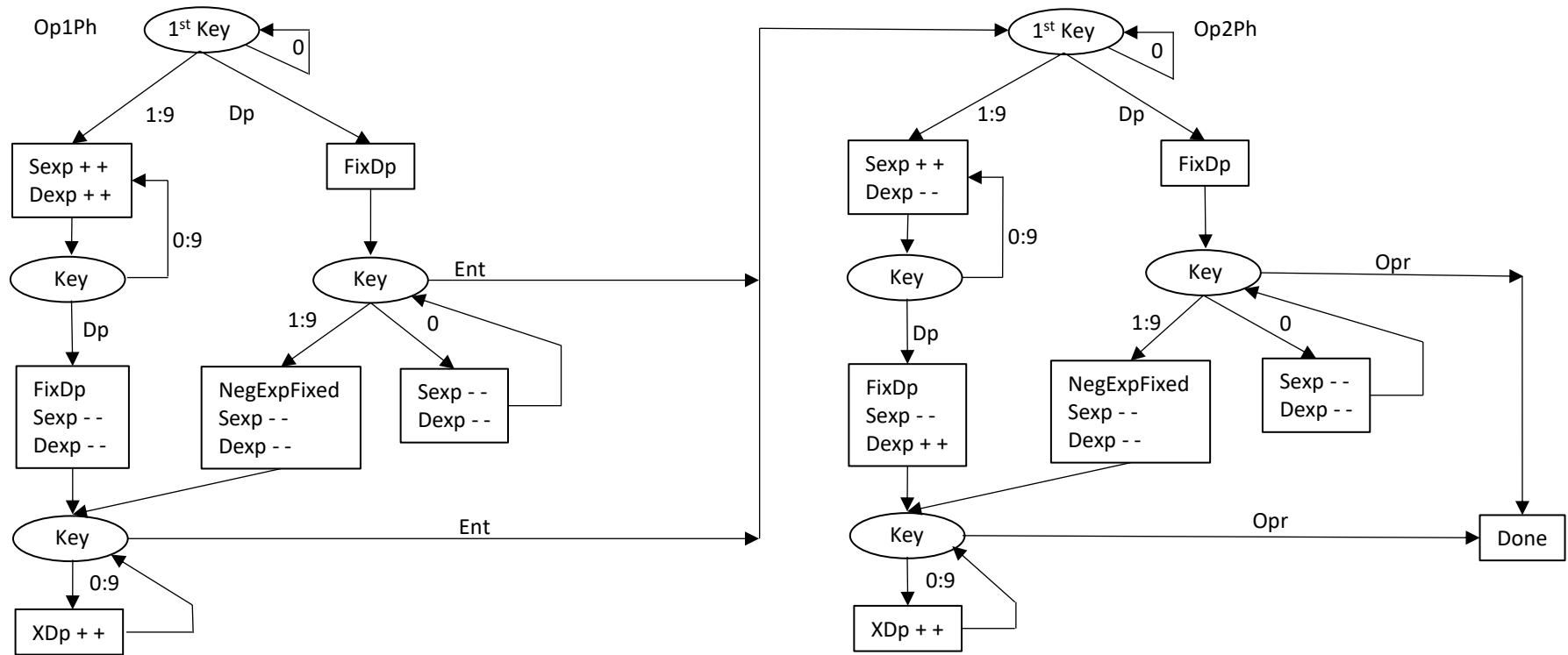
Mult



Div



Exponent Computation (XDp Placement)



Add/Sub:

If $Dexp \leq 0$, then $XDp = XDp(Op2)$ else $XDp = XDp(Op2) + Dexp$.

Mult

Pos Exp: XDp placement is initialized by $Sexp$. Incremented once (XDp shifted right one digit) if the final Mult loop leaves a non-zero accumulation of CarryOs in X9.

Neg Exp: XDp placed in left most position. $Sexp$ number of leading zeros right shifted into MSD of product, losing right most digits for each shift.

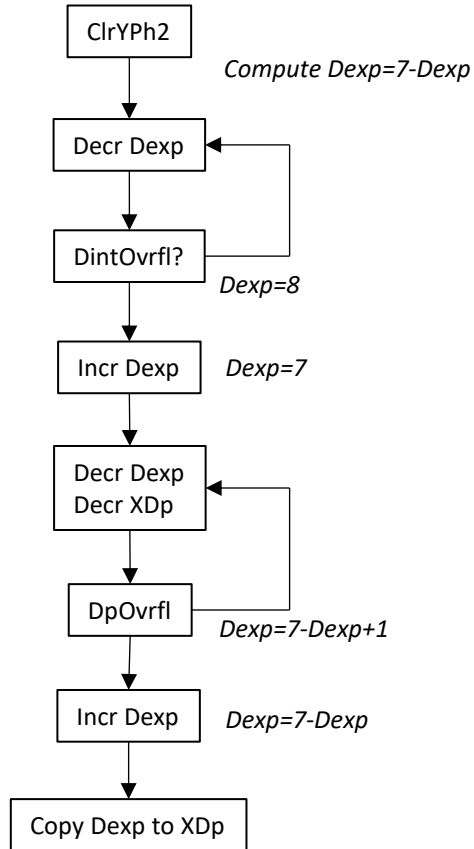
Div:

Pos Exp: XDp placement is initialized by $Dexp$. Decremented once (XDp shifted left one digit) if the minimum number of dividend digits needed to subtract the divisor with a positive result is one more than the divisor digits.

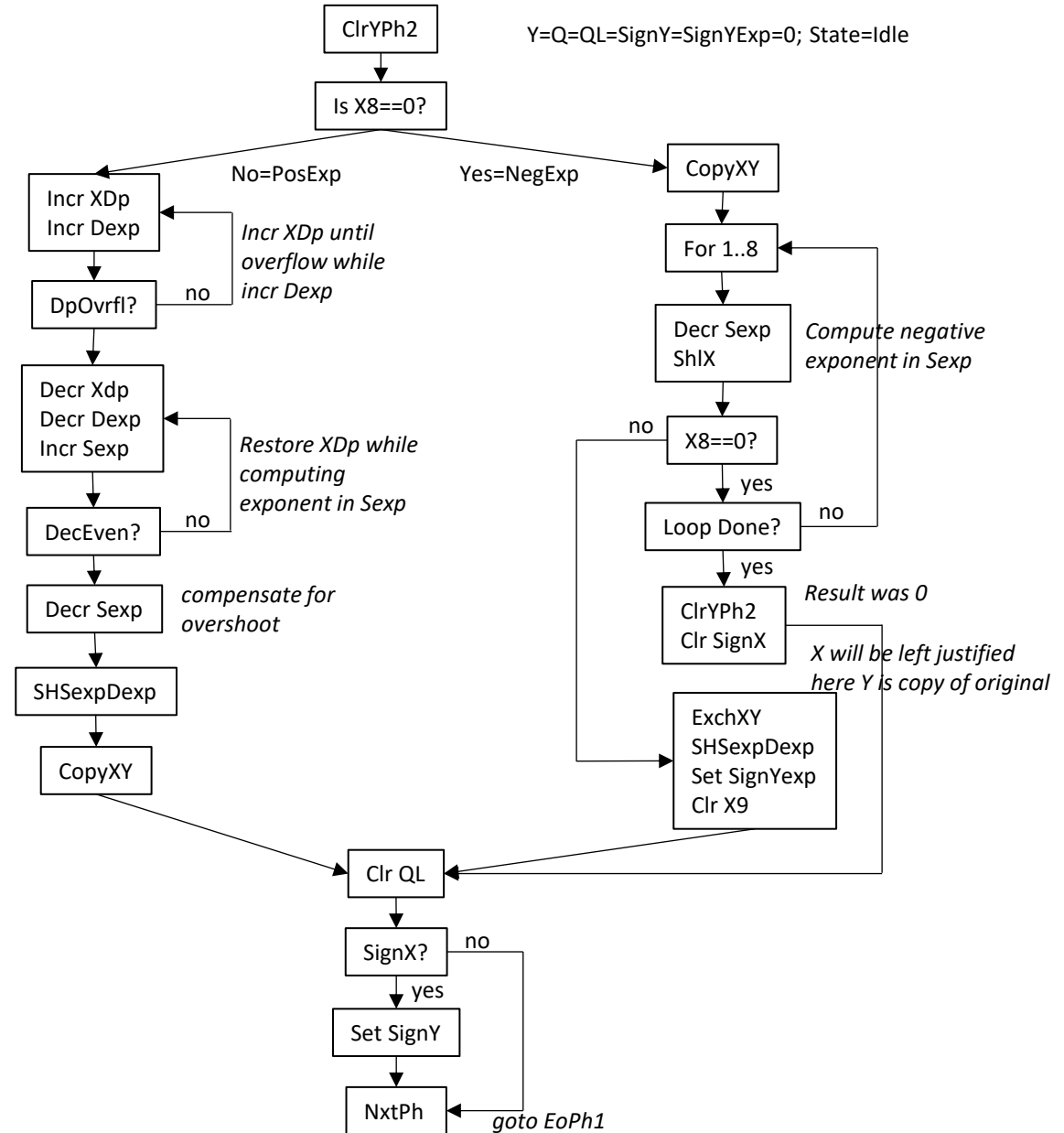
NegExp: XDp placed in left most position. $Dexp$ number of leading zeros right shifted into MSD, losing rightmost digits for each shift.

Operation Chaining (Post processing the result of an operation to support chaining)

Invert XDP



Y=Q=QL=SignY=SignYExp=0; State=Idle



9+0.0000001=9.0000001
99999999+.1=99999999
99+0.0000001=99

Addition Overflow and Underflow

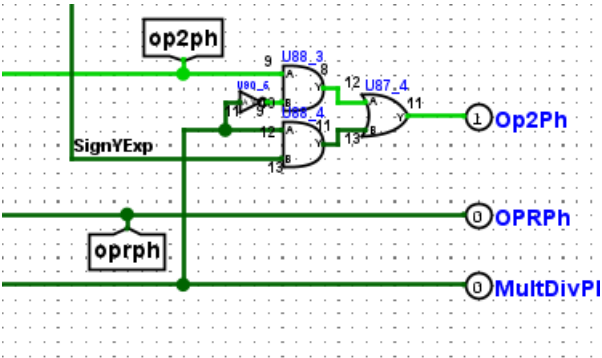
DPFirst=1
Dintovrfl=1

0000	0	
0001	1	-15
0010	2	-14
0011	3	-13
0100	4	-12
0101	5	-11
0110	6	-10
0111	7	-9
1000	8	-8
1001	9	-7
1010	10	-6
1011	11	-5
1100	12	-4
1101	13	-3
1110	14	-2
1111	15	-1

Xexp(+), Yexp(+)										Xexp(-), Yexp(+)							
OP2(X)		10000000	1000000	100000	10000	1000	100	10	1	0.1	0.01	0.001	0.0001	0.00001	0.000001	0.0000001	
OP1 (Y)		exp	7	6	5	4	3	2	1	0	-1	-2	-3	-4	-5	-6	-7
10000000		7	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1000000		6	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13
100000		5	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
10000		4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11
1000		3	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10
100		2	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9
10		1	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8
1		0	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
Xexp(+), Yexp(-)										Xexp(-), Yexp(-)							
0.1		-1	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
0.01		-2	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5
0.001		-3	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4
0.0001		-4	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
0.00001		-5	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
0.000001		-6	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1
0.0000001		-7	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0

Dexp Counter values for All Op1 and Op2 Combinations

Add Overflow (DintOvrfl=1)(SignYexp=0)(SignXexp=0)
OR
(DintOvrfl=1)(SignYexp=1)(SignXexp=1)
Overflow happens when Op1 and Op2 are both 8-digit integers and the sum results in a final carry out digit 8.



Op2(X) is DPFirst AND DintOvrfl
Add Underflow (DintOvrfl=1)(SignYexp=0)(SignXexp=1)
Use Yreg
Add Underflow (DintOvrfl=1)(SignYexp=1)(SignXexp=0)
Use Xreg
Op2(X) is NOT DintOvrfl AND NOT DPFirst

Because SignYexp is only available to Mult and Div (due to arch oversight) I can't Use these expressions to detect underflow.

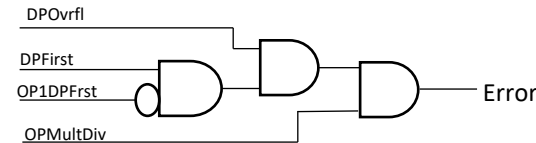
Multiplication and Division Overflows and Underflows

		OP2																																	
A div C...	0.0000001	0.0000001	0.000001	0.000001	0.00001	0.00001	0.0001	0.0001	0.001	0.001	0.01	0.01	0.1	0.1	0	0	1	1	10	10	100	100	1000	1000	10000	10000	100000	100000	1000000	1000000	10000000	10000000			
	-7	-7	-6	-6	-5	-5	-4	-4	-3	-3	-2	-2	-1	-1			0	0	1	1	10	10	100	100	1000	1000	10000	10000	100000	100000	1000000	1000000	10000000	10000000	
	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp			Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	
0.0000001	-7	0	-14	-1	-13	-2	-12	-3	-11	-4	-10	-5	-9	-6	-8			-7	-7	-8	-6	-9	-5	-10	-4	-11	-3	-12	-2	-13	-1	-14	0	-13	1
0.000001	-6	1	-13	0	-12	-1	-11	-2	-10	-3	-9	-4	-8	-5	-7			-6	-6	-7	-5	-8	-4	-9	-3	-10	-2	-11	-1	-12	0	-13	1	-12	2
0.00001	-5	2	-12	1	-11	0	-10	-1	-9	-2	-8	-3	-7	-4	-6			-5	-5	-6	-4	-7	-3	-8	-2	-9	-1	-10	0	-11	1	-12	2	-11	3
0.0001	-4	3	-11	2	-10	1	-9	0	-8	-1	-7	-2	-6	-3	-5			-4	-4	-5	-3	-6	-2	-7	-1	-8	0	-9	1	-10	2	-11	3	-10	4
0.001	-3	4	-10	3	-9	2	-8	1	-7	0	-6	-1	-5	-2	-4			-3	-3	-4	-2	-5	-1	-6	0	-7	1	-8	2	-9	3	-10	4	-9	5
0.01	-2	5	-9	4	-8	3	-7	2	-6	1	-5	0	-4	-1	-3			-2	-2	-3	-1	-4	0	-5	1	-6	2	-7	3	-8	4	-9	5	-8	6
0.1	-1	6	-8	5	-7	4	-6	3	-5	2	-4	1	-3	0	-2			-1	-1	-2	0	-3	1	-4	2	-5	3	-6	4	-7	5	-8	6	-7	7
OP1	0																																		
	1	0	7	-7	6	-6	5	-5	4	-4	3	-3	2	-2	1	-1			0	0	-1	1	-2	2	-3	3	-4	4	-5	5	-6	6	-7	7	
	10	1	8	-6	7	-5	6	-4	5	-3	4	-2	3	-1	2	0			1	1	0	2	-1	3	-2	4	-3	5	-4	6	-5	7	-6	8	
	100	2	9	-5	8	-4	7	-3	6	-2	5	-1	4	0	3	1			2	2	1	3	0	4	-1	5	-2	6	-3	7	-4	8	-5	9	
	1000	3	10	-4	9	-3	8	-2	7	-1	6	0	5	1	4	2			3	3	2	4	1	5	0	6	-1	7	-2	8	-3	9	-4	10	
	10000	4	11	-3	10	-2	9	-1	8	0	7	1	6	2	5	3			4	4	3	5	2	6	1	7	0	8	-1	9	-2	10	-3	11	
	100000	5	12	-2	11	-1	10	0	9	1	8	2	7	3	6	4			5	5	4	6	3	7	2	8	1	9	0	10	-1	11	-2	12	
	1000000	6	13	-1	12	0	11	1	10	2	9	3	8	4	7	5			6	6	5	7	4	8	3	9	2	10	1	11	0	12	-1	13	
	10000000	7	14	0	13	1	12	2	11	3	10	4	9	5	8	6			7	7	6	8	5	9	4	10	3	11	2	12	1	13	0	14	

Dexp and Sexp Counter Values From All Combinations of Op1 and Op2

Division Overflows
Anytime Dexp is positive number greater than 7
Within the positive exp1, negative exp2 quadrant

$(DPOvrfl=1)(SignYexp=0)(SignXexp=1)$

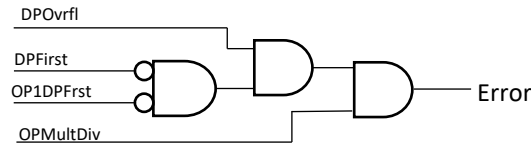


Div Underflows

$(DPOvrfl=1)(SignYexp=1)(SignXexp=0)$

Mult Overflows
Anytime Sexp is positive number greater than 7
Within the positive exp1, positive exp2 quadrant

$(DPOvrfl=1)(SignYexp=SignXexp=0)$



Mult Underflows

$(DPOvrfl=1)(SignYexp=SignXexp=1)$